

JOUL, Volume 3

Supplemental Information

**Projecting the Future Levelized Cost
of Electricity Storage Technologies**

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Supplemental Tables

Table S1. Review of 27 unique-purpose electricity storage services and allocation to core services based on similar technical requirements

Application	Description	Alternative name	Core application
Wholesale arbitrage	Purchase power in low-price periods and sell in high price periods on the energy wholesale market ¹	Electric Energy Time-shift	Energy arbitrage
Retail arbitrage	Purchase power in low-price periods and sell in high price periods on the energy retail market ¹	End-consumer arbitrage	Energy arbitrage
Regulating reserve	Automatically correct the continuous, fast, frequent changes in load or generation within the shortest applicable market interval ²	Frequency regulation, Frequency control	Primary response
Primary reserve	Automatically stabilise frequency after rare, sudden change in load or generation ²	Primary contingency reserve, Frequency response	Primary response
Following reserve	Manually correct anticipated imbalances between load and generation ²	Load following, Balancing reserve	Secondary response
Secondary reserve – spinning	Automatically return frequency to nominal after rare, sudden change in load or generation with operating generator ³	Spinning reserve	Secondary response
Secondary reserve – non-spinning	Automatically return frequency to nominal after rare, sudden change in load or generation with non-operating generator ³	Secondary contingency reserve, Non-spinning reserve	Secondary response
Ramping reserve	Manually correct for unexpected, severe and infrequent changes in load or generation that are not instantaneous ²	-	Secondary response
Renewables integration - uncertainty	Change and optimise output from variable supply resources when generation is out of line with forecasts ⁴	Correct for forecasting inaccuracy, Renewables capacity firming	Secondary response
Tertiary reserve	Automatically replace primary and secondary contingency reserve ²	Tertiary contingency reserve, Supplemental / Replacement reserve	Tertiary response
Peaker replacement	Ensure availability of sufficient generation capacity at all times ¹	Electric supply / System capacity, Capacity mechanism, Microgrid	Peaker replacement
Black start	Restore power plant operations after network outage without external power supply ¹	-	Black start
Seasonal storage	Compensate longer-term supply disruption or seasonal variability in supply and demand ⁵	-	Seasonal storage
Transmission upgrade deferral	Defer transmission infrastructure upgrades required when peak power flows exceed existing capacity ¹	Transmission support, Network efficiency	T&D deferral
Distribution upgrade deferral	Defer distribution infrastructure upgrades required when peak power flows exceed existing capacity ¹	Distribution substation, Network efficiency	T&D deferral
Transmission congestion relief	Avoid risk of overloading existing infrastructure that could lead to re-dispatch and local price differences ¹	Transmission support, Network efficiency	Congestion management

Bill management	Purchase power in low-price periods and use during high-price periods ¹	Energy management, Retail ToU charges	Bill management
Demand charge reduction – R	Reduce demand supplied by the network during periods of highest retail network charges ¹	Peak reduction, Retail demand charges	Bill management
Demand charge reduction – D	Reduce demand supplied by the network during periods of highest distribution network cost ⁴	Peak reduction, Red zone management	Bill management
Demand charge reduction – T	Reducing demand supplied by the network during periods of highest transmission network cost ⁴	Peak reduction, Triad avoidance, Transmission access charges	Bill management
Renewable energy self-consumption	Minimise export of renewable electricity and increase self-consumption to maximise financial benefits ⁶	-	Bill management
Power quality	Protect on-site load against short-duration power loss or variations in voltage or frequency ¹	-	Power quality
Power reliability	Fill gap between variable resource and demand ⁵	Off-grid, On-site power	Power reliability
Backup power	Provide sustained power during total loss of power from source utility ¹	Home backup, Emergency supply, Resiliency	Power reliability
Renewables integration - variability	Change and optimise output from variable supply resources to mitigate output changes and match supply with demand ⁵	Off-peak storage, Variable resource integration, On-site generation shifting	Power reliability
Voltage support	Maintain voltage levels across networks via reactive power supply/reduction ¹	-	-
VAR support	Maintain voltage levels across transmission network via reactive power supply/reduction ³	-	-

Table S2. Technical requirements for electricity storage applications

Application	Size (MW)	Duration (hours)	Cycles (per year)	Response Time (seconds)
Energy arbitrage	0.001-2,000 ^{1,5}	1-24 ^{1,5}	50-400 ^{1,3,5}	>10 ⁵
Primary response	1-2,000 ⁵	0.02-1 ^{1,5}	250-15,000 ^{1,5}	<10 ²
Secondary response	10-2,000 ⁵	0.25-24 ^{1,5}	20-10,500 ^{1,5}	>10 ²
Tertiary response	5-1,000 ^{2,5}	>1.5 ²	20-50 ¹	>10 ²
Peaker replacement	1-500 ¹	2-6 ¹	5-100 ¹	>10 ⁵
Black start	0.1-400 ⁵	0.25-4 ^{1,5}	1-20 ^{1,5}	>10 ⁵
Seasonal storage	500-2,000 ⁵	24-2000 ⁵	1-5 ⁵	>10 ⁵
T&D upgrade deferral	1-500 ⁵	2-8 ¹	10-500 ^{1,5}	>10 ⁵
Congestion management	1-500 ^{1,5}	1-4 ¹	50-500 ^{1,5}	>10 ⁵
Bill management	0.001-10 ¹	1-6 ¹	50-500 ¹	>10 ⁵
Power quality	0.05-10 ^{1,3}	0.003-0.5 ^{1,3}	10-200 ¹	<10 ¹
Power reliability	0.001-10 ³	2-10 ³	50-400 ³	>10 ¹

Note: *Cycles* refers to full equivalent charge-discharge cycles. Superscripts refer to references.

Table S3. Electricity storage technology performance characteristics

Technology	Power range (MW)	Discharge (hours)	Cycle life (# cycles)	Response time (seconds)
Pumped hydro	10-5,000 ⁷	1-24 ⁷	20,000-50,000 ⁷	> 10 ⁷
Compressed air	5-400 ⁷	1-24 ⁷	>13,000 ⁷	> 10 ⁷
Flywheel	0.01-20 ⁸	< 0.5 ⁷	20,000-225,000 ^{1,7}	< 10 ⁷
Lead-acid	0.005-100 ¹	0.25-10 ¹	< 5,500 ¹	< 10 ⁷
Lithium-ion	0.001-35 ⁹	0.25-5 ¹	2,000-3,500 ⁹	< 10 ⁷
Sodium-sulphur	0.05-50 ^{7,9}	0.0167-8 ^{1,9}	2,500-4,500 ⁹	< 10 ¹⁰
Redox-flow	0.02-50 ¹	0.0167-10 ⁷	5,000-13,000 ^{7,9}	< 10 ⁷
Hydrogen	0.3-500 ⁷	0.0167-24 ⁷	<20,000 ⁷	< 10 ⁷
Supercapacitor	<4 ¹¹	<1 ¹¹	>100,000 ⁷	< 10 ⁷

Note: *Cycles* refers to full equivalent charge-discharge cycles. Superscripts refer to references.

Table S4. Technology input parameters for 2015 (standard deviation)

			Pumped hydro	Compressed air	Flywheel	Lithium-ion	Sodium-sulphur	Lead-acid	Vanadium redox-flow	Hydrogen	Super-capacitor
Investment cost - Power	\$/kW	C _P	1129 (45%)	871 (35%)	641 (17%)	678 (17%)	657 (27%)	675 (23%)	829 (21%)	5417 (48%)	296 (31%)
Investment cost - Energy	\$/kWh	C _E	80 (63%)	39 (58%)	5399 (67%)	802 (24%)	738 (12%)	471 (38%)	760 (17%)	31 (60%)	13560 (19%)
Operation cost - Power	\$/kW-yr	C _{P-OM}	8 (26%)	4 (23%)	7 (8%)	10 (35%)	11 (50%)	8 (31%)	12 (52%)	46 (30%)	0 (0%)
Operation cost - Energy	\$/MWh	C _{E-OM}	1 (60%)	4 (60%)	2 (60%)	3 (60%)	3 (60%)	1 (60%)	1 (60%)	0 (60%)	0 (60%)
Replacement cost	\$/kW	C _{P-r}	116 (5%)	93 (5%)	199 (44%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1637 (48%)	0 (0%)
Replacement interval	cycles	C _{Cr}	7300	1460	22500	3250	4098	1225	8272	6388	69320
End-of-life cost	%	F _{EOl}	0%	0%	0%	0%	0%	0%	0%	0%	0%
Discount rate	%	DR	8%	8%	8%	8%	8%	8%	8%	8%	8%
Round-trip efficiency	%	η _{RT}	78% (9%)	44% (16%)	88% (3%)	86% (7%)	81% (6%)	84% (0%)	73% (9%)	40% (13%)	91% (6%)
Self-discharge	%/day	η _{self,idle}	0%	0%	480%	0%	20%	0%	0%	1%	30%
Lifetime (100% DoD)	cycles	C _{Clife}	33250 (43%)	16250 (20%)	143402 (30%)	3250 (38%)	4098 (29%)	1225 (35%)	8272 (13%)	20000 (0%)	300000 (67%)
Shelf life	years	T _{shelf}	55 (9%)	30 (33%)	18 (14%)	13 (38%)	14 (20%)	10 (50%)	13 (20%)	18 (14%)	14 (33%)
Response time	seconds		>10	>10	<10	<10	<10	<10	<10	<10	<10
Time degradation	%/year	T _{deg}	0.4%	0.7%	1.3%	1.7%	1.6%	2.2%	1.7%	1.3%	1.6%
Cycle degradation	%/cycle	C _{ycdeg}	0.0007%	0.0014%	0.0002%	0.0069%	0.0054%	0.0182%	0.0027%	0.0011%	0.0001%
Construction time	years	T _c	3	2	1	1	1	1	1	1	1
Sources			1,7,12–15	1,7,12–14,16,17	1,3,7,12–14	7,9,13,14,18	1,7,9,13,14,18	1,7,12–14,19,20	1,7,9,13,14	7,13,14,21–25	7,12–14

Note: Cycles refers to full equivalent charge-discharge cycles.

Table S5. Technology cycle life relative to depth-of-discharge (DoD)

Depth-of-Discharge	Pumped hydro	Compressed air	Flywheel	Lithium-ion	Sodium-sulphur	Lead-acid	Vanadium redox-flow	Hydrogen	Super-capacitor
100%	33,250	16,250	143,402	3,250	4,098	1,225	8,272	20,000	300,000
90%	33,250	16,250	143,402	4,875	4,131	1,336	8,272	20,000	300,000
80%	33,250	16,250	143,402	6,297	4,193	1,501	8,272	20,000	300,000
70%	33,250	16,250	143,402	8,531	4,592	1,763	8,272	20,000	300,000
60%	33,250	16,250	143,402	10,766	5,299	2,074	8,272	20,000	300,000
50%	33,250	16,250	143,402	14,219	6,006	2,598	8,272	20,000	300,000
40%	33,250	16,250	143,402	18,586	7,050	3,194	8,272	20,000	300,000
30%	33,250	16,250	143,402	24,984	8,516	4,211	8,272	20,000	300,000
20%	33,250	16,250	143,402	35,953	10,654	6,316	8,272	20,000	300,000
10%	33,250	16,250	143,402	60,734	21,325	13,183	8,272	20,000	300,000
Source				26	27	19			

Note: Cycles refers to full equivalent charge-discharge cycles.

Table S6. Optimal depth-of-discharge (if applicable)

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Table S7. Combined standard deviations for investment cost parameters

Investment cost - Power	2015	2020	2025	2030	2035	2040	2045	2050
Pumped hydro	45%	45%	45%	45%	45%	45%	46%	46%
Compressed air	35%	35%	35%	35%	35%	36%	36%	36%
Flywheel	17%	17%	19%	23%	29%	32%	33%	34%
Lithium-ion	17%	28%	45%	59%	65%	67%	67%	68%
Sodium-sulphur	27%	28%	29%	32%	36%	39%	40%	41%
Lead-acid	23%	24%	24%	24%	24%	24%	24%	24%
Vanadium redox-flow	21%	37%	51%	59%	61%	60%	58%	57%
Hydrogen	48%	48%	49%	51%	53%	55%	56%	57%
Supercapacitor	31%	31%	32%	35%	39%	41%	42%	43%
Investment cost - Energy	2015	2020	2025	2030	2035	2040	2045	2050
Pumped hydro	63%	63%	63%	63%	63%	63%	64%	64%
Compressed air	58%	58%	58%	58%	59%	59%	59%	59%
Flywheel	67%	67%	67%	69%	71%	72%	73%	73%
Lithium-ion	24%	33%	48%	61%	67%	69%	70%	70%
Sodium-sulphur	12%	13%	15%	20%	26%	30%	31%	32%
Lead-acid	38%	38%	39%	38%	38%	38%	38%	38%
Vanadium redox-flow	17%	35%	49%	58%	60%	58%	57%	56%
Hydrogen	60%	60%	60%	62%	64%	66%	66%	67%
Supercapacitor	19%	19%	20%	24%	30%	33%	34%	35%

Table S8. Investment cost projections relative to 2015 with forecast uncertainty

	2015	2020	2025	2030	2035	2040	2045	2050	Comment
Pumped hydro	100% (0%)	100% (0%)	100% (1%)	100% (3%)	101% (6%)	101% (8%)	102% (10%)	102% (12%)	Original
Compressed air	100% (0%)	100% (0%)	100% (1%)	100% (3%)	101% (6%)	101% (8%)	102% (10%)	102% (12%)	Same as pumped hydro
Flywheel	100% (0%)	84% (3%)	66% (6%)	53% (8%)	44% (10%)	39% (11%)	36% (10%)	33% (10%)	Same as hydrogen
Lithium-ion	100% (0%)	55% (12%)	34% (14%)	23% (13%)	18% (12%)	16% (10%)	15% (10%)	14% (9%)	Original
Sodium-sulphur	100% (0%)	84% (3%)	66% (6%)	53% (8%)	44% (10%)	39% (11%)	36% (10%)	33% (10%)	Same as hydrogen
Lead-acid	100% (0%)	80% (5%)	68% (6%)	63% (5%)	61% (5%)	59% (4%)	59% (4%)	58% (5%)	Original
Vanadium redox-flow	100% (0%)	49% (15%)	34% (16%)	26% (14%)	21% (12%)	19% (11%)	18% (10%)	17% (9%)	Original
Hydrogen	100% (0%)	84% (3%)	66% (6%)	53% (8%)	44% (10%)	39% (11%)	36% (10%)	33% (10%)	Original
Supercapacitor	100% (0%)	84% (3%)	66% (6%)	53% (8%)	44% (10%)	39% (11%)	36% (10%)	33% (10%)	Same as hydrogen

Supplemental Figures

Figure S1. Core application with modelled technical requirements

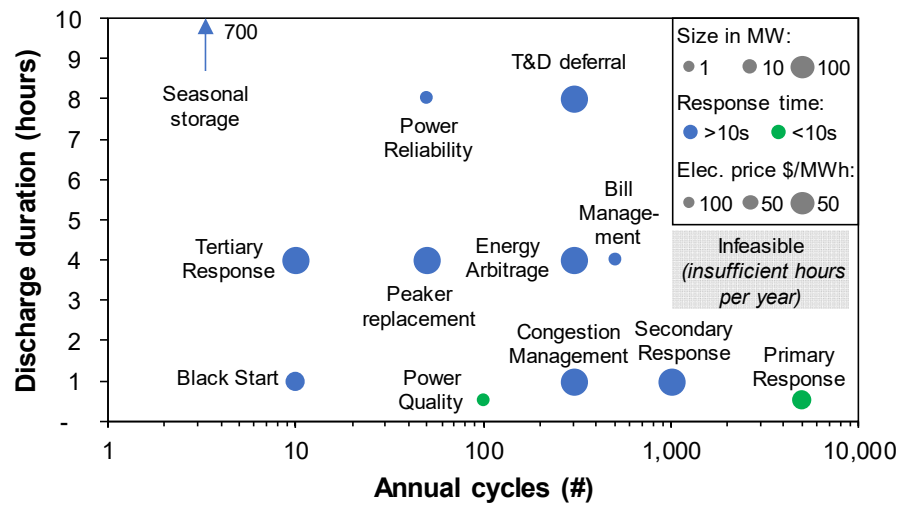


Figure S2. Updated experience curve data set

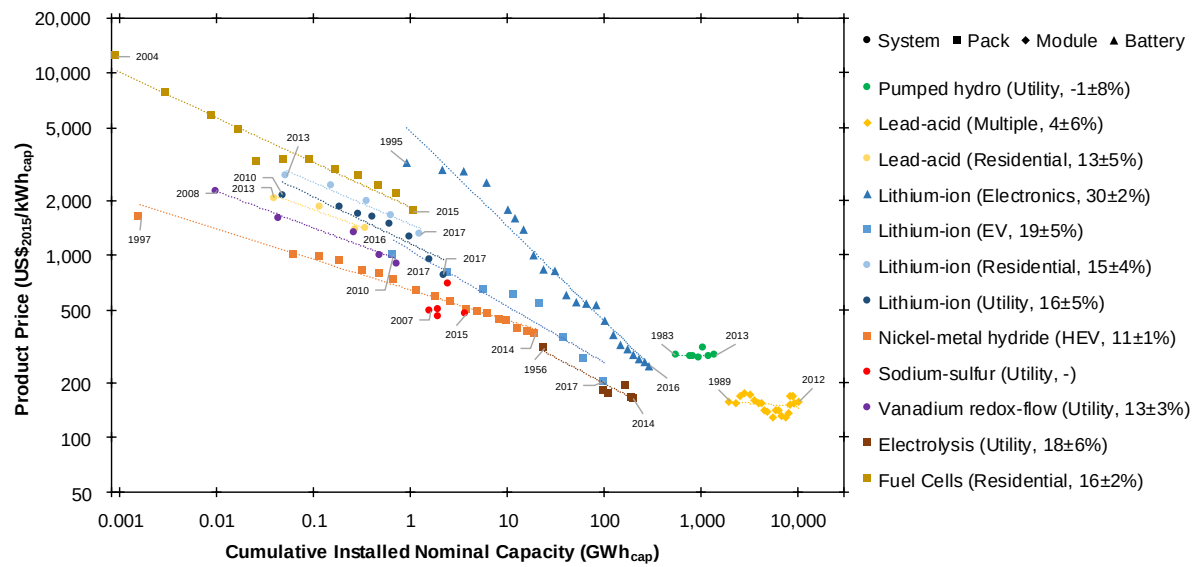
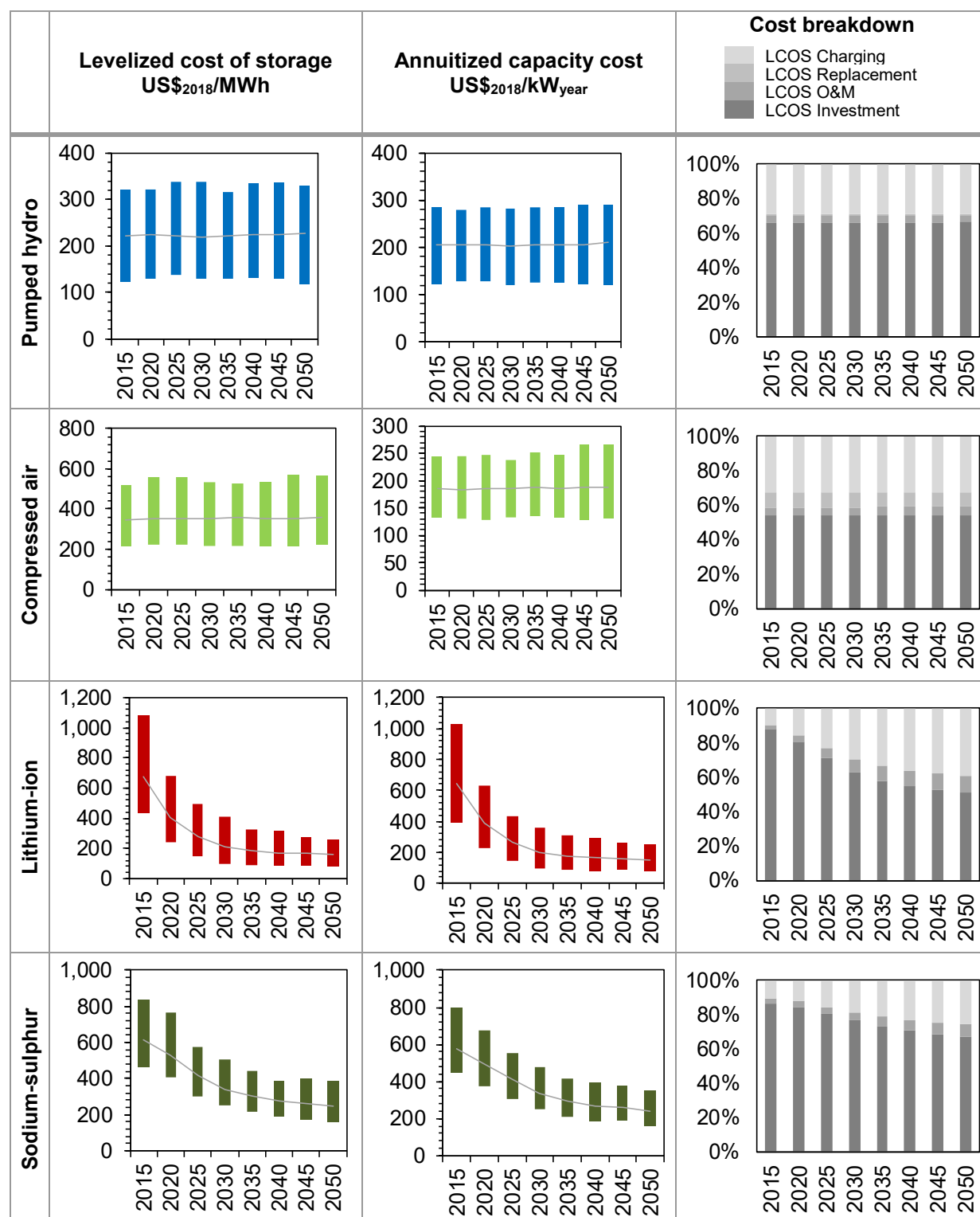
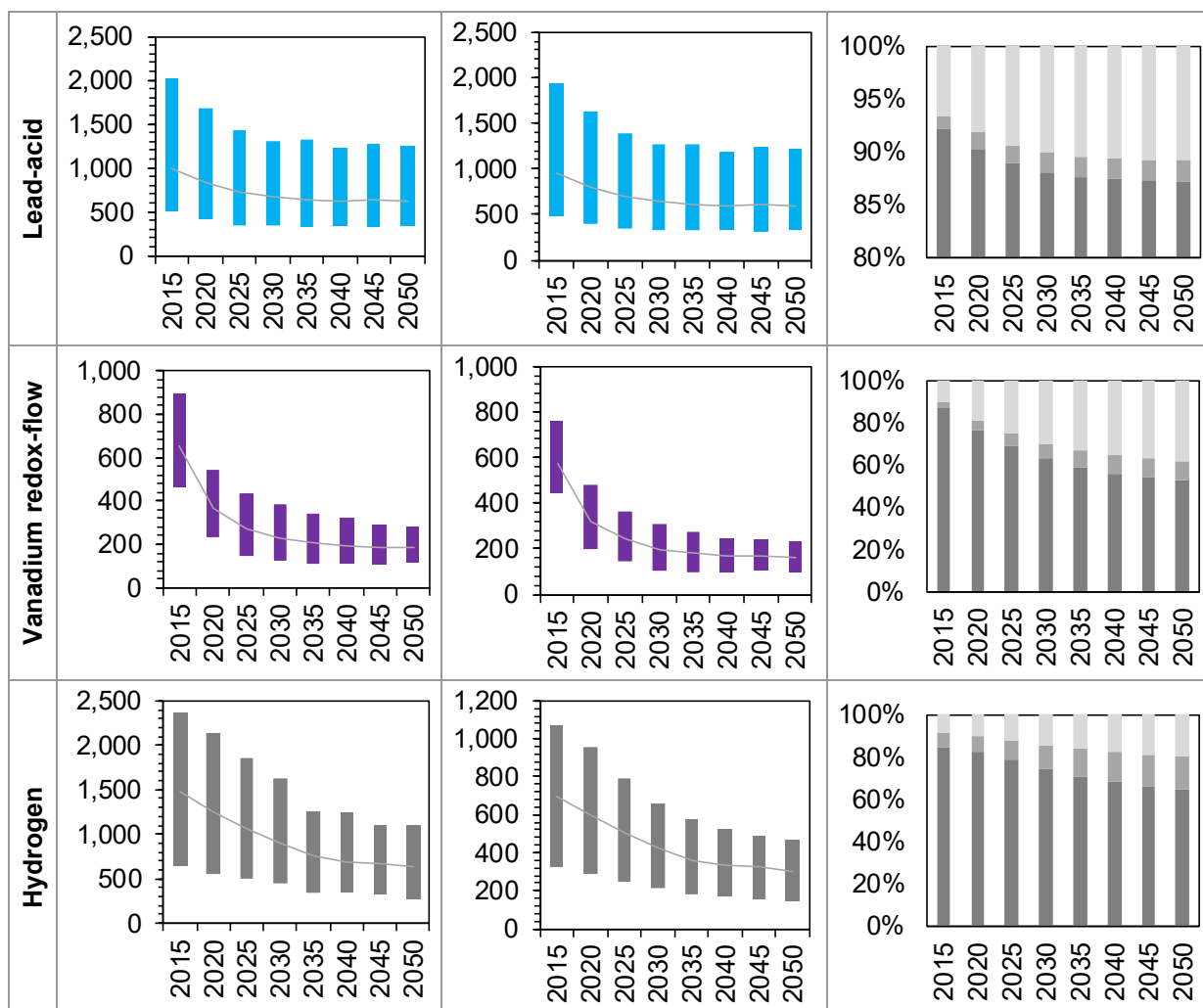


Figure S2 – Updated experience curve data set based on Schmidt, 2017²⁸. All lithium-ion and redox-flow data sets have been updated to include 2016 and/or 2017 data. The data set including future price projections can be downloaded²⁹.

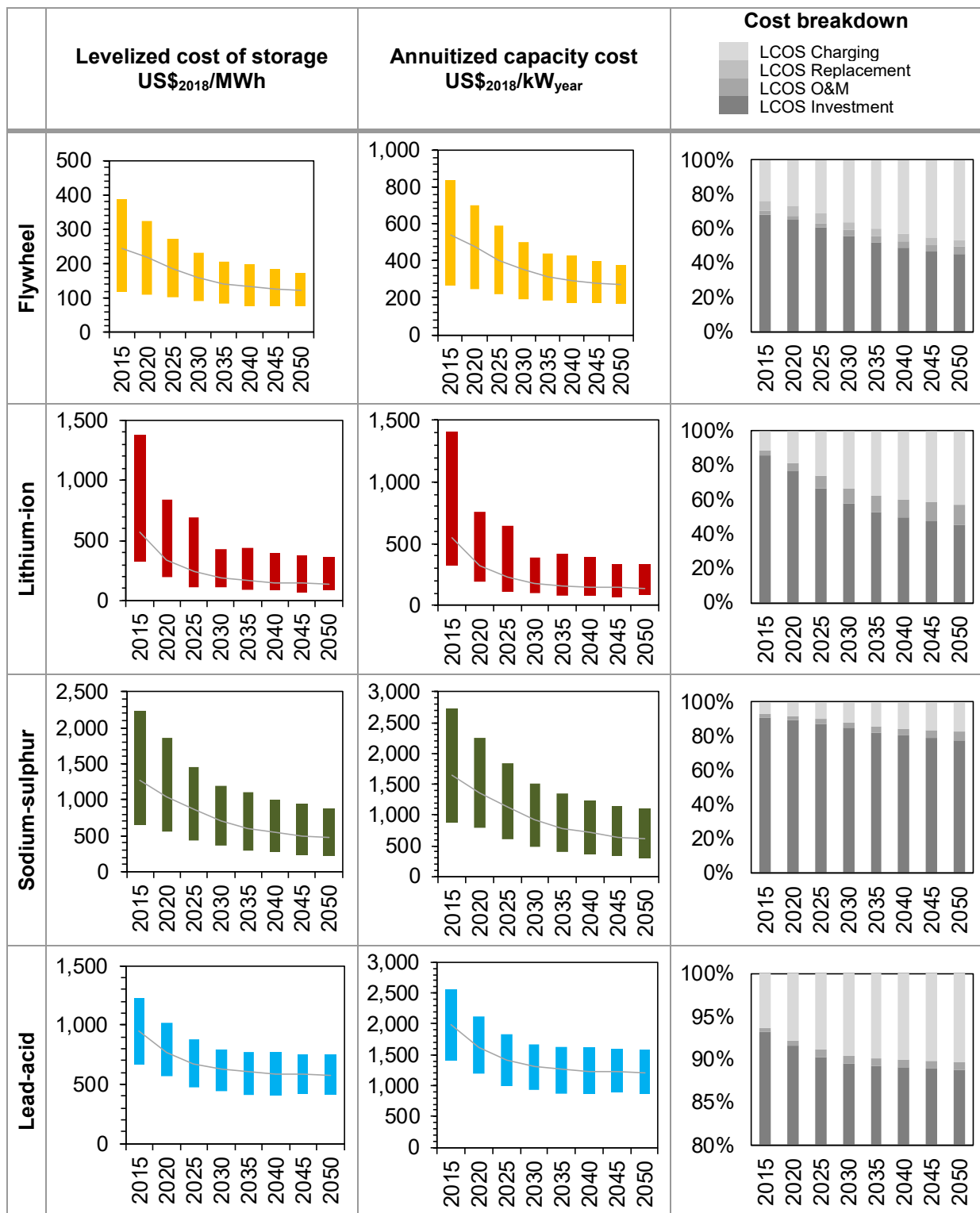
Figure S3. LCOS projections and breakdown for 9 technologies in 12 applications

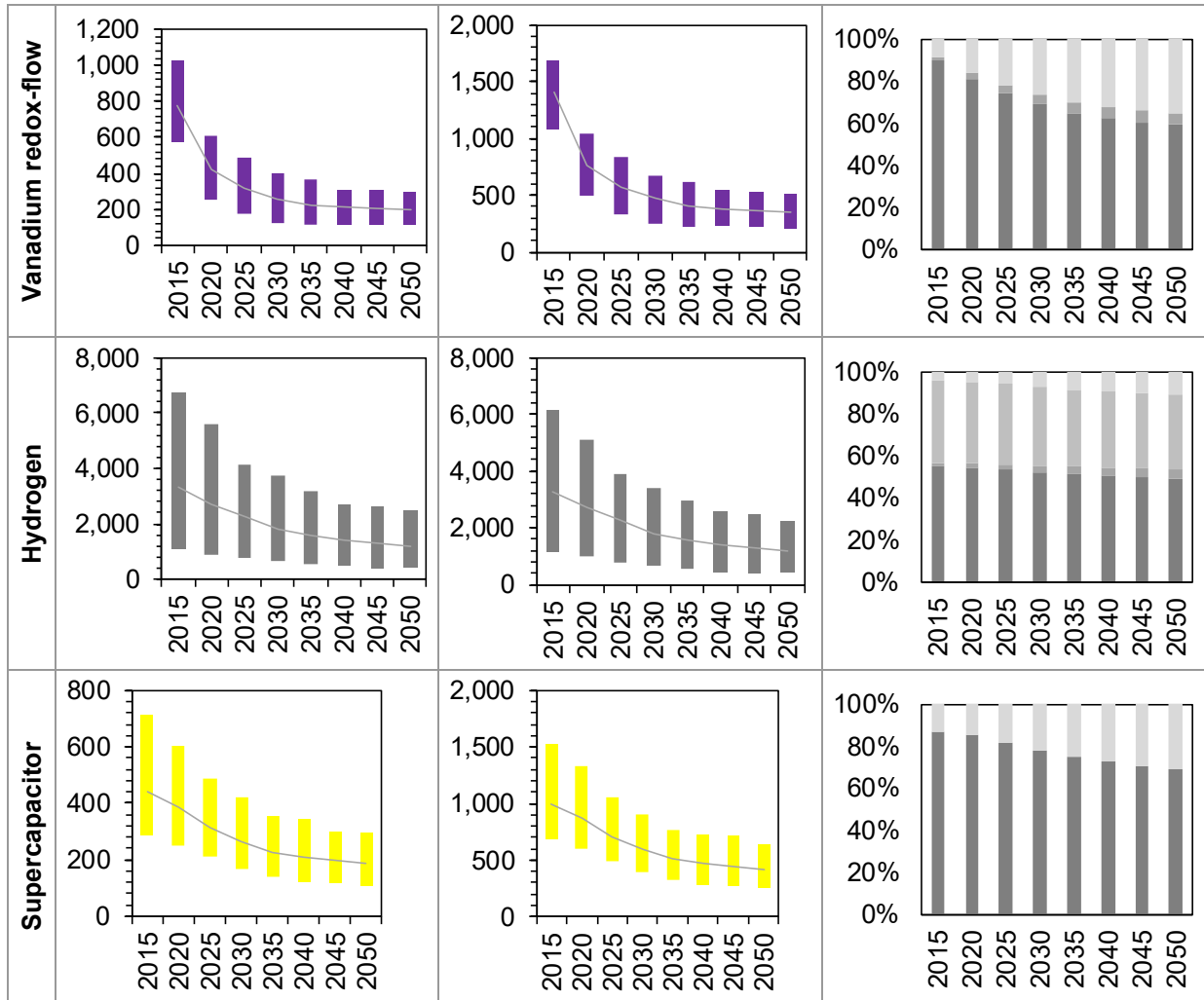
Application: Energy Arbitrage



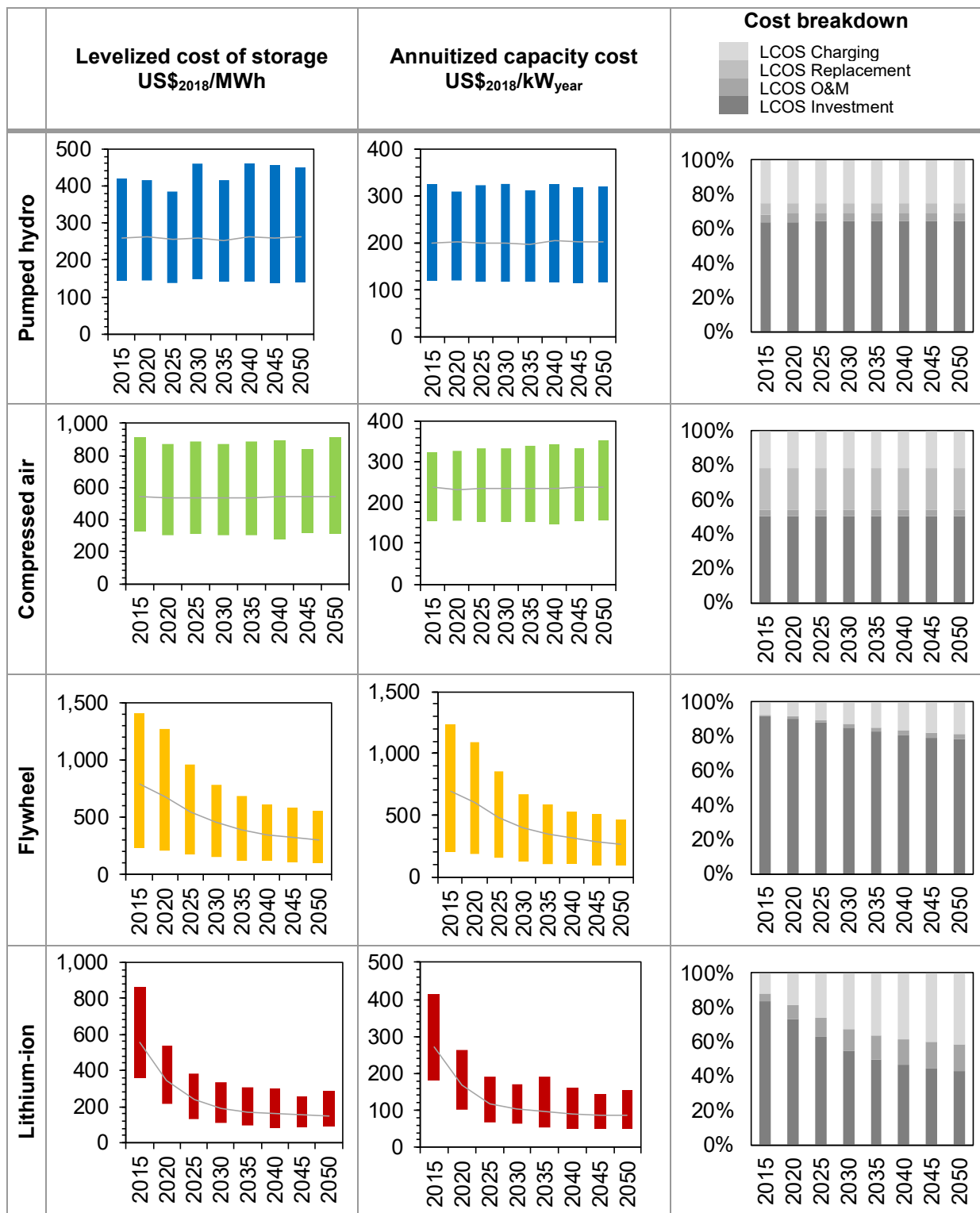


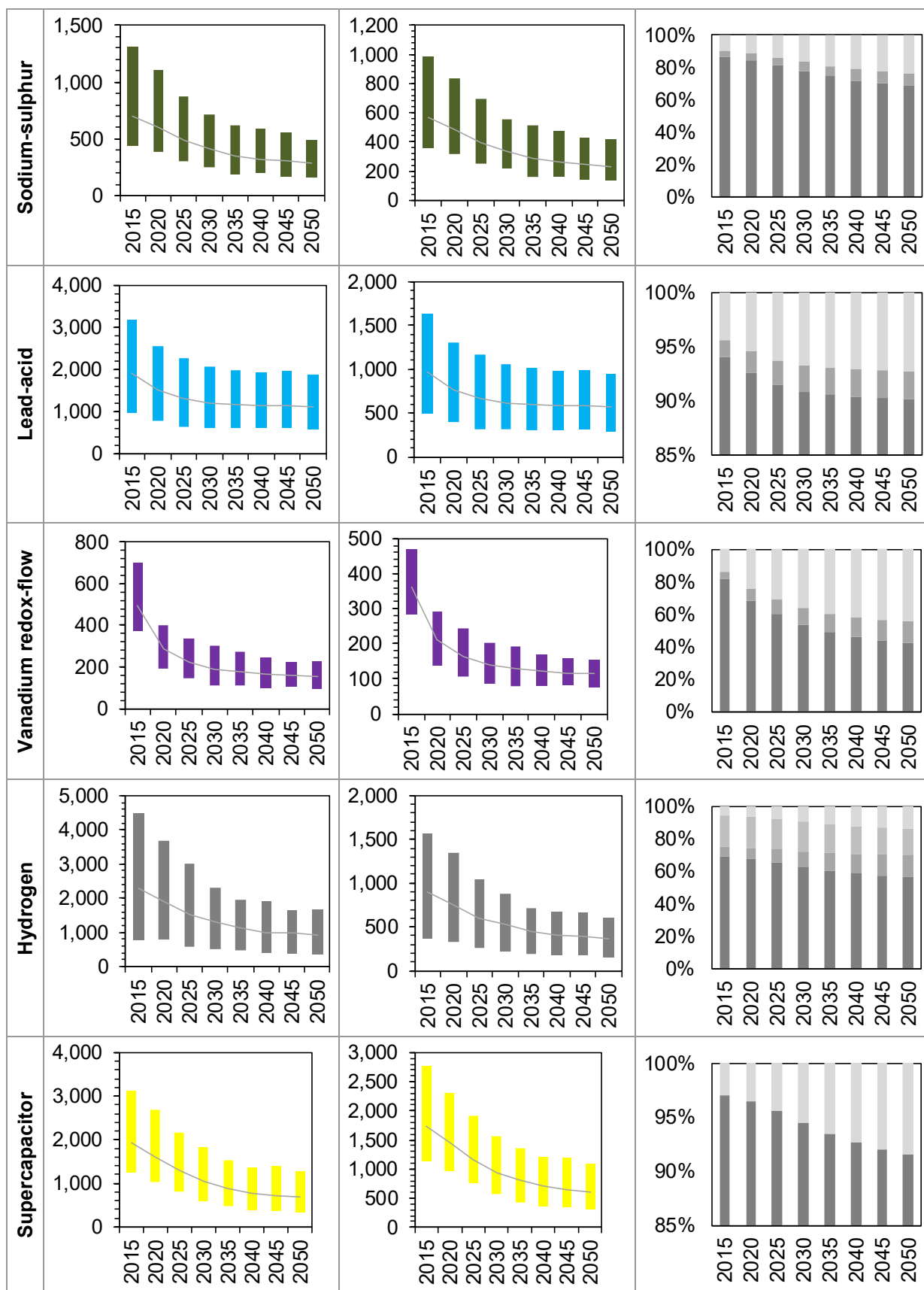
Application: Primary Response



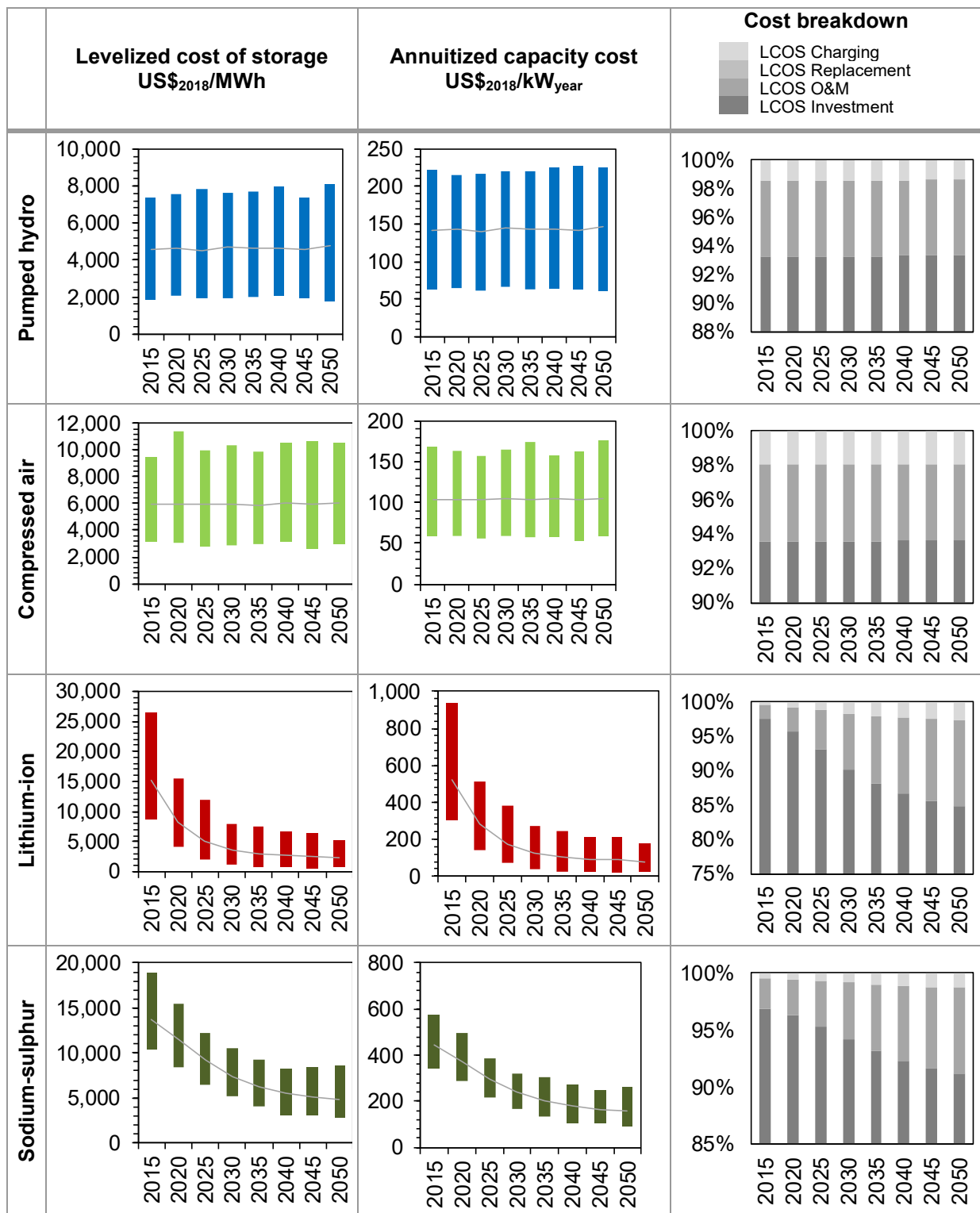


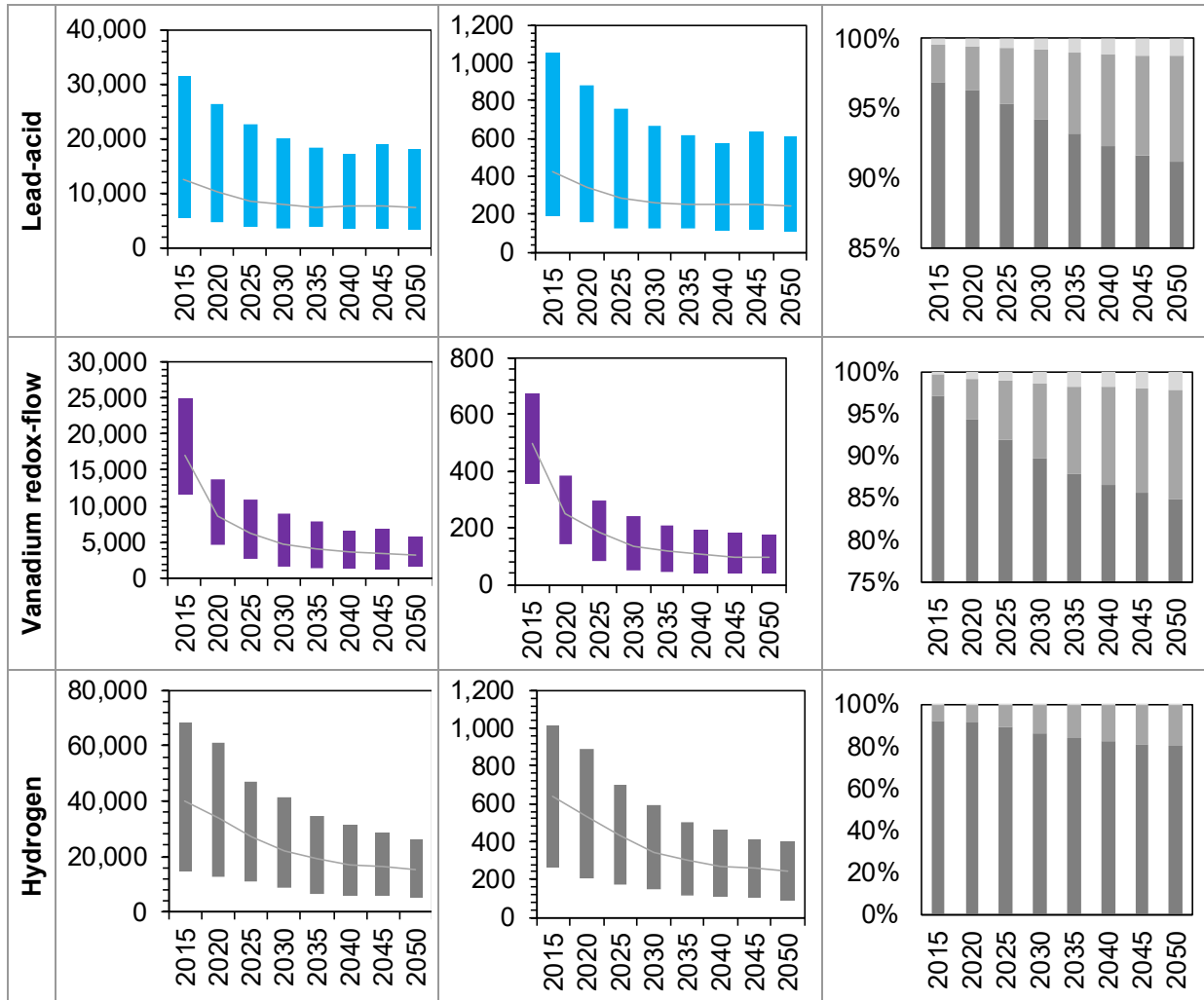
Application: Secondary Response



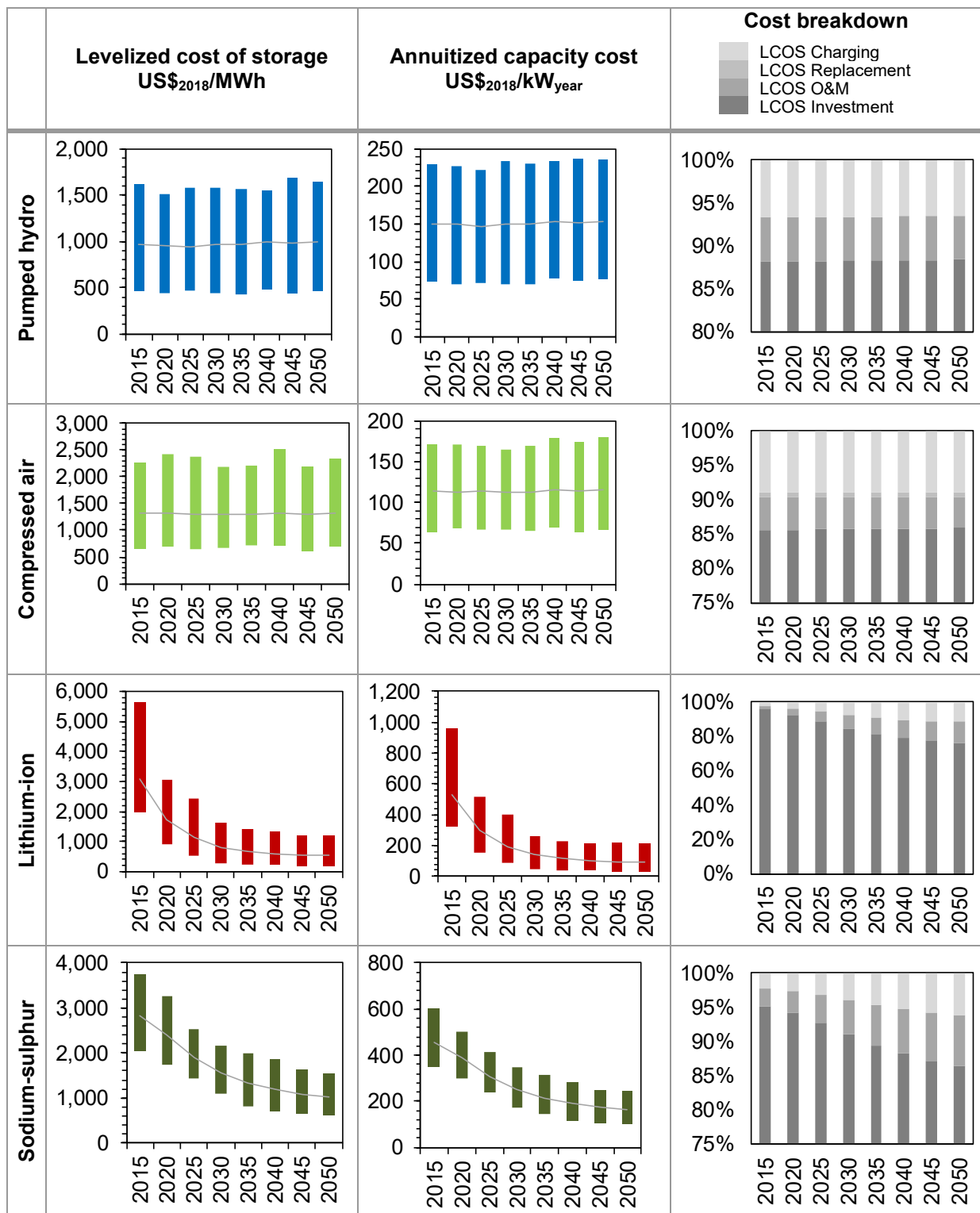


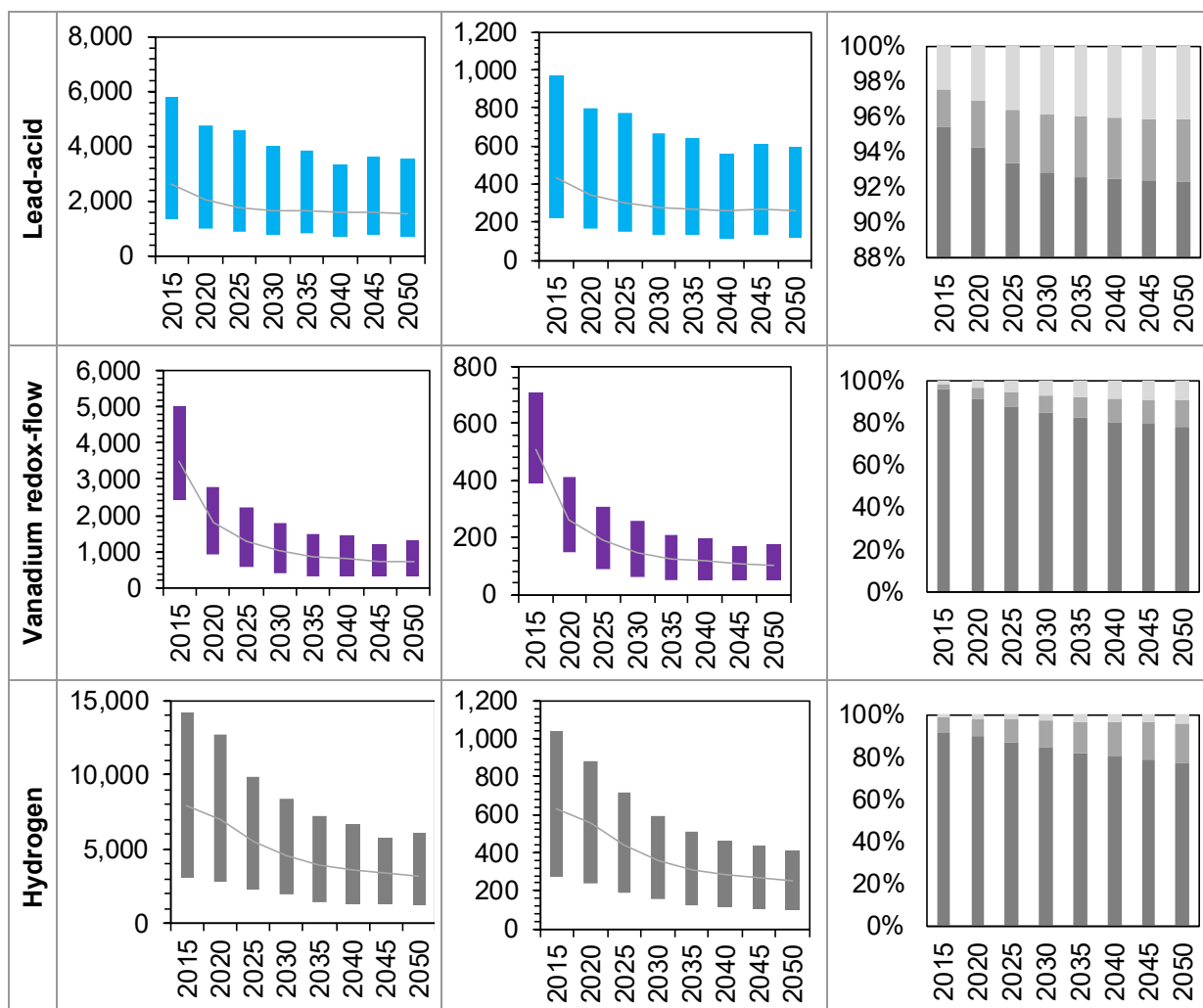
Application: Tertiary Response



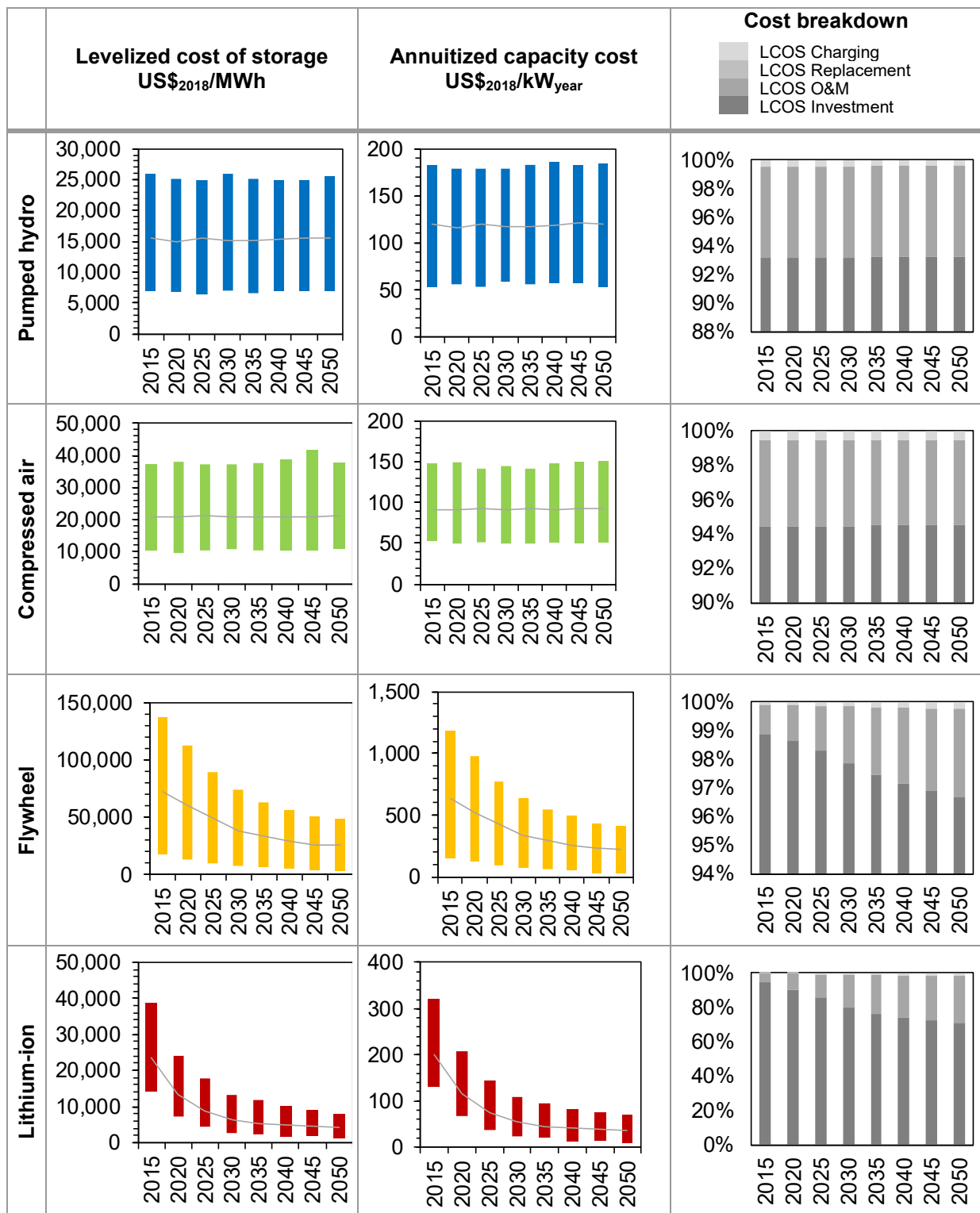


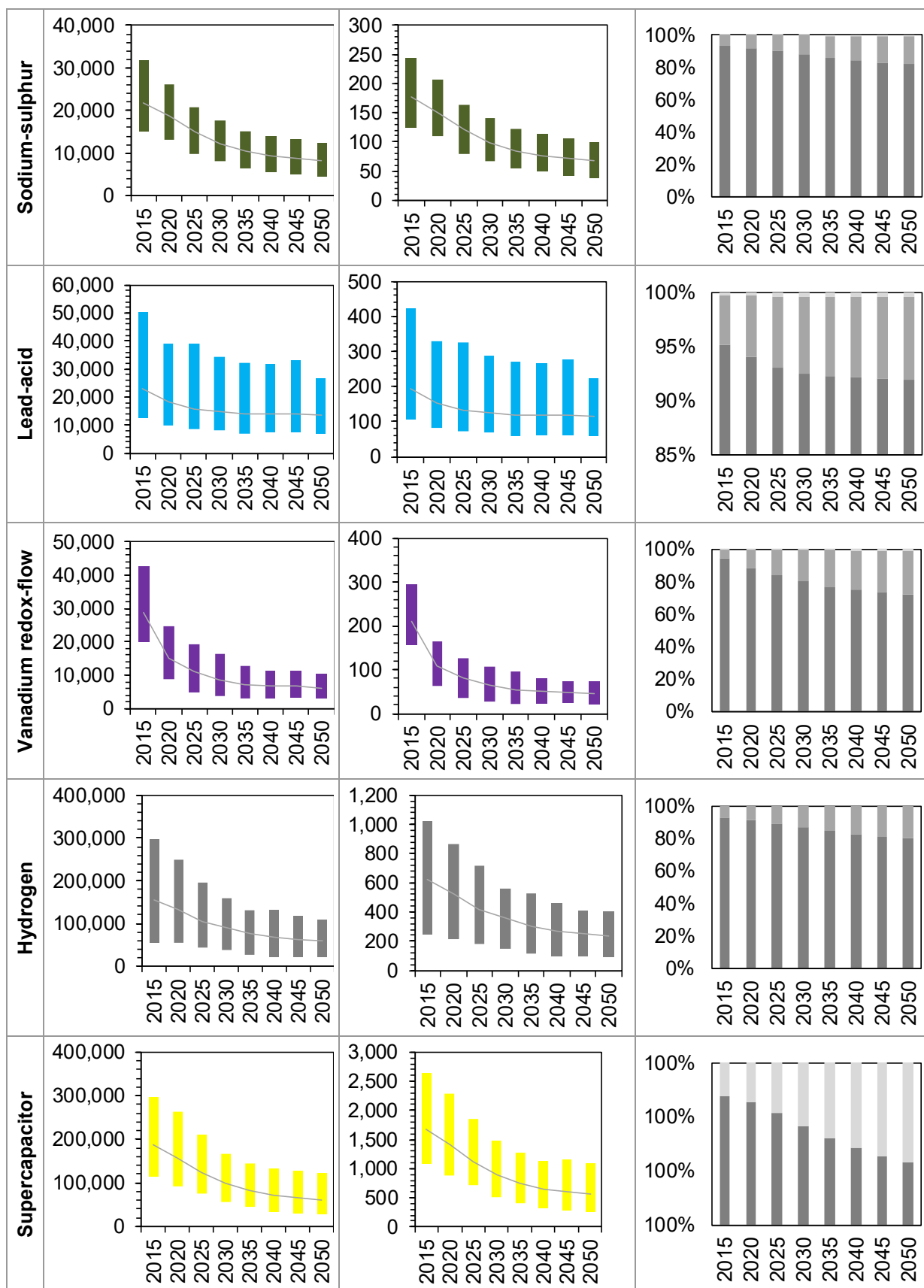
Application: Peaker Replacement



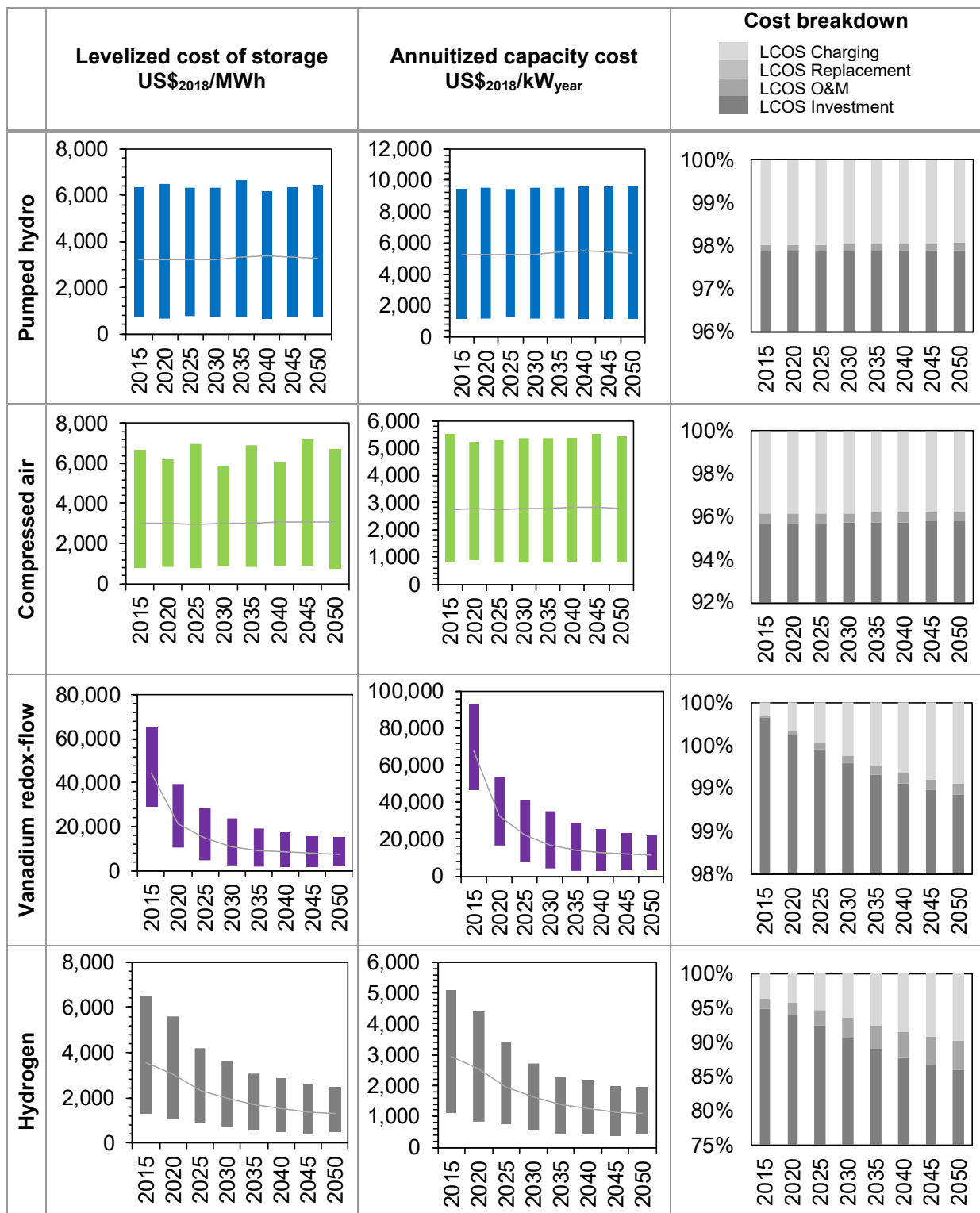


Application: Black Start

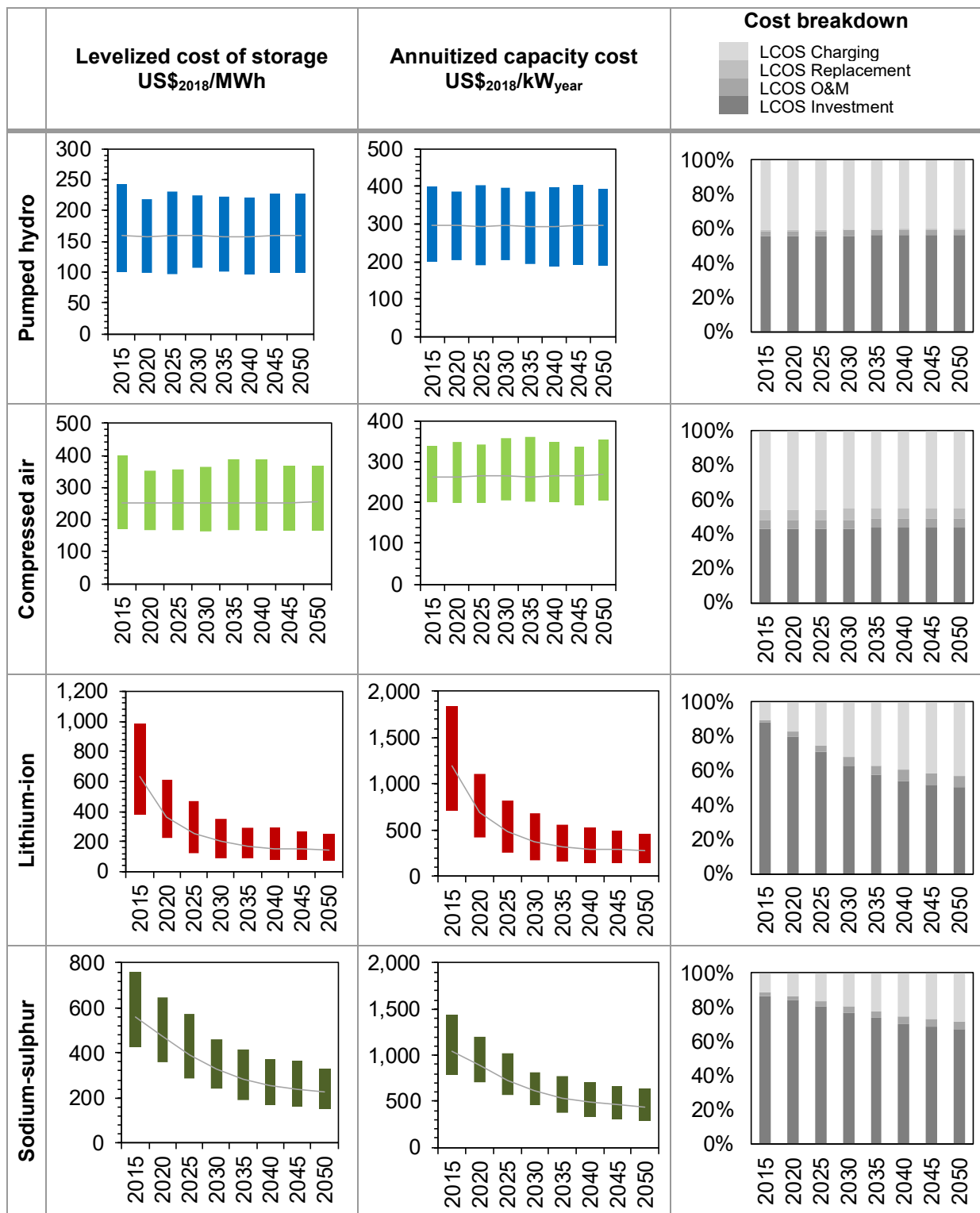


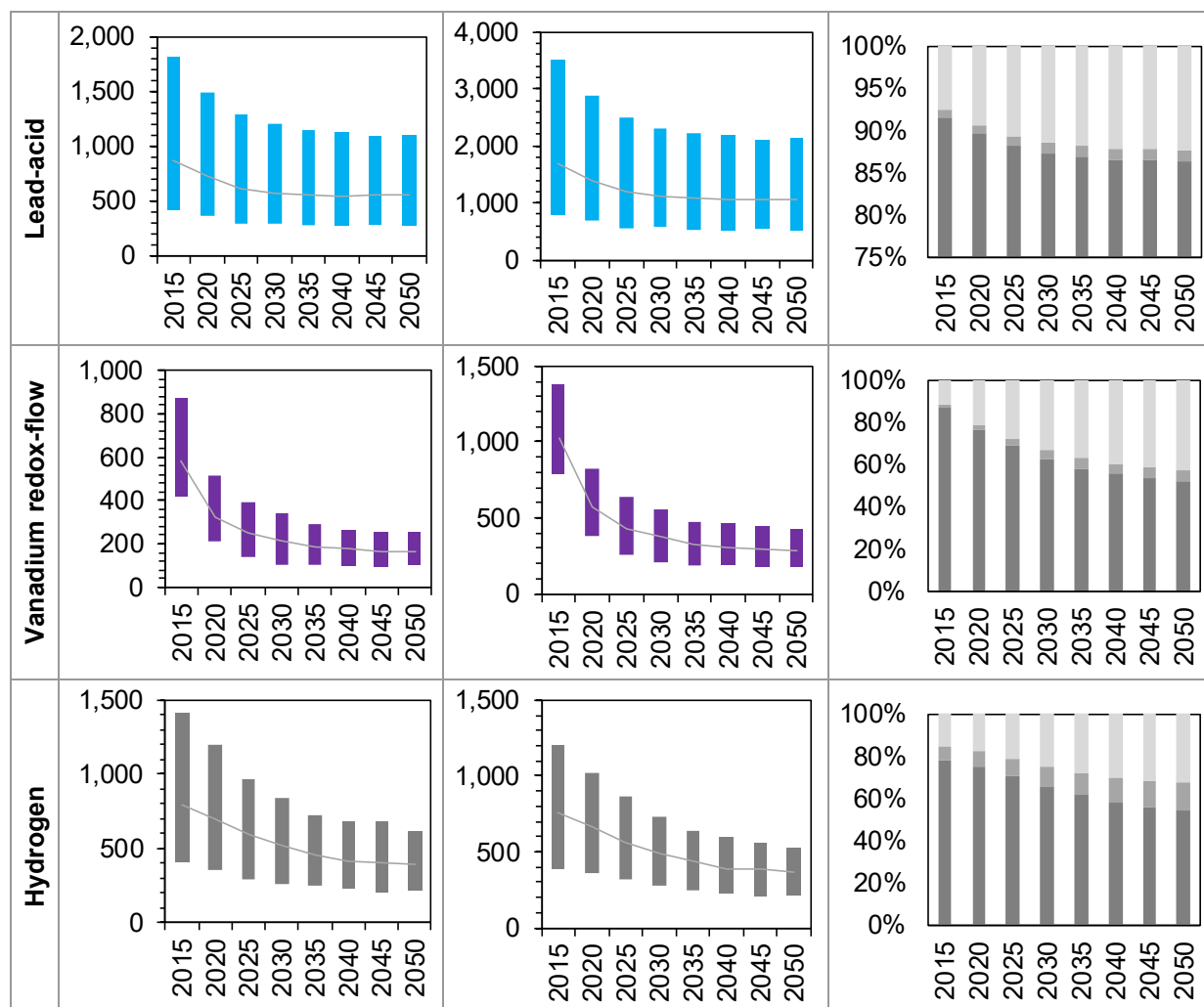


Application: Seasonal Storage

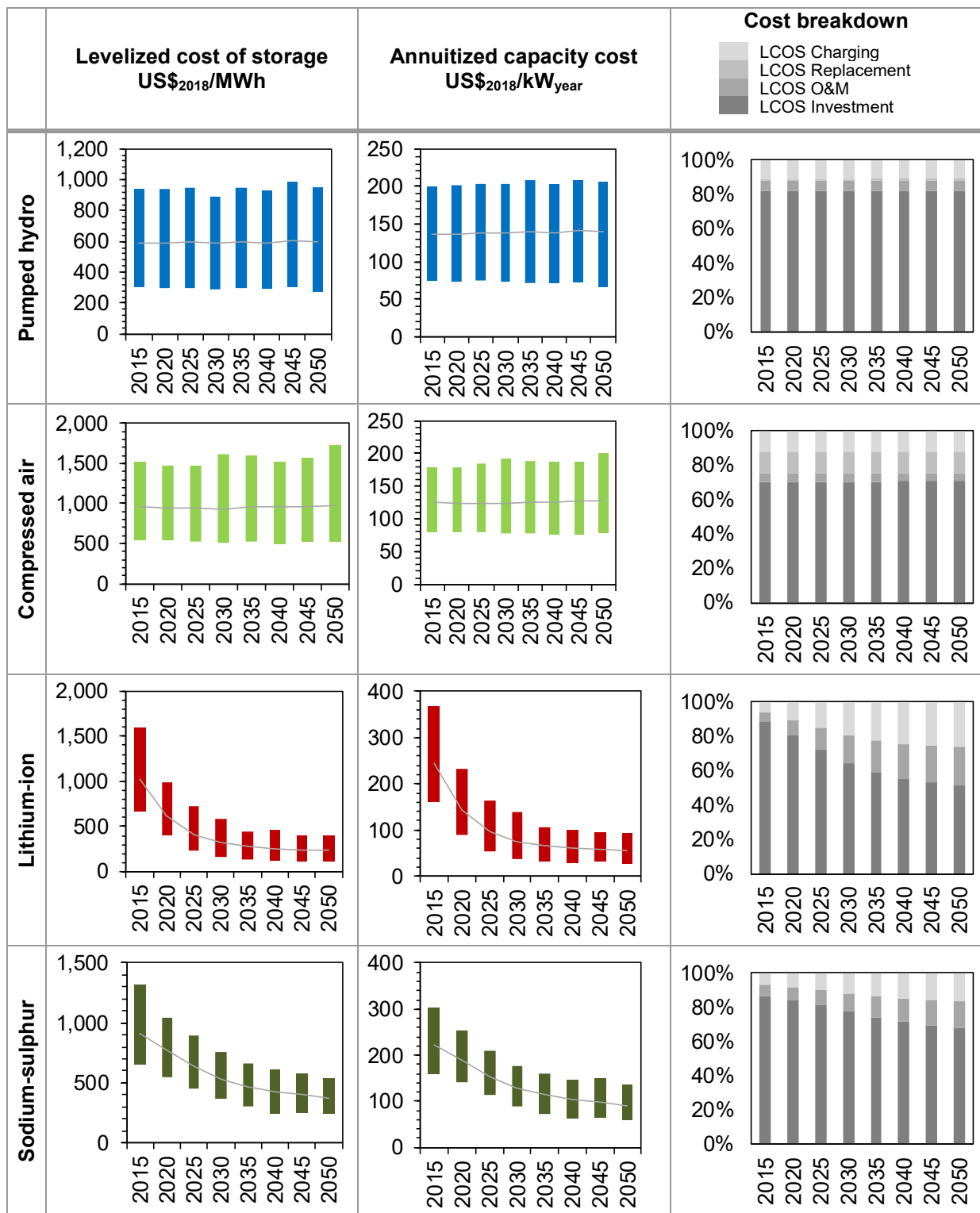


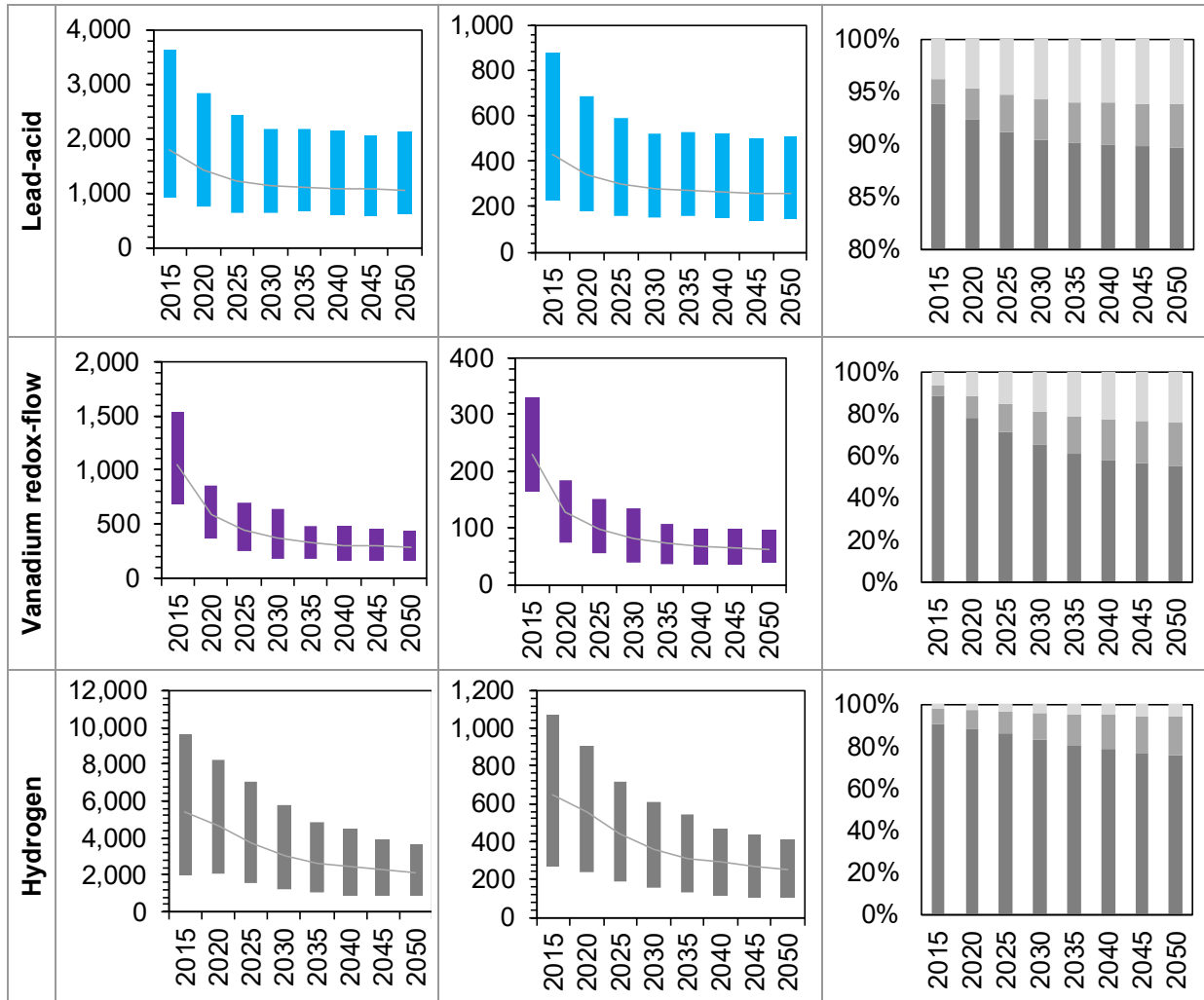
Application: T&D upgrade deferral



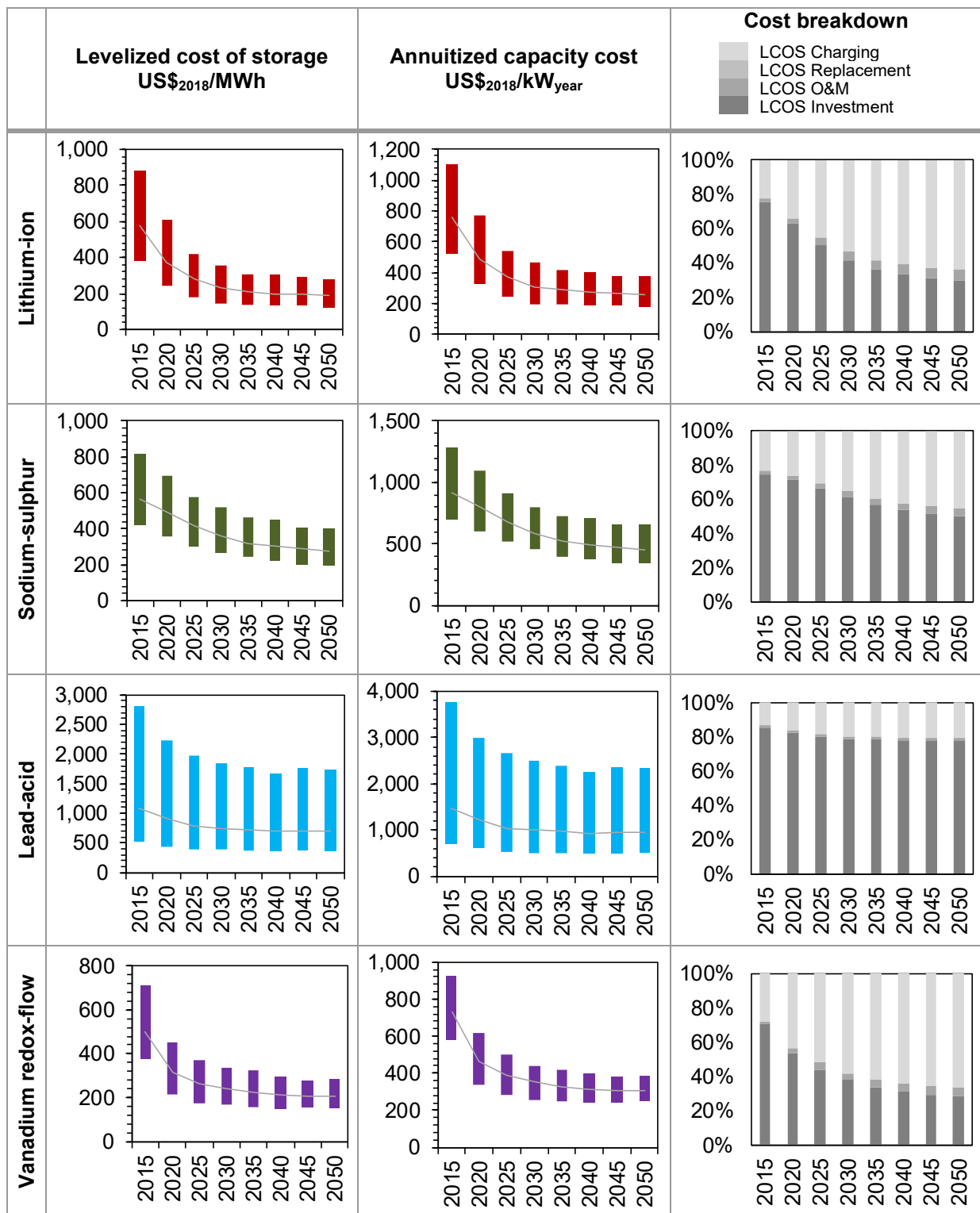


Application: Congestion Management

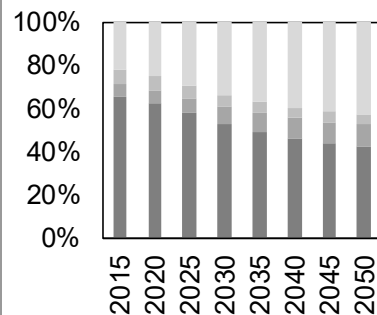
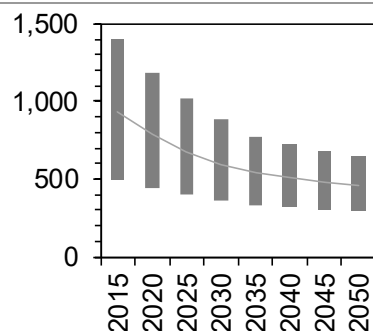
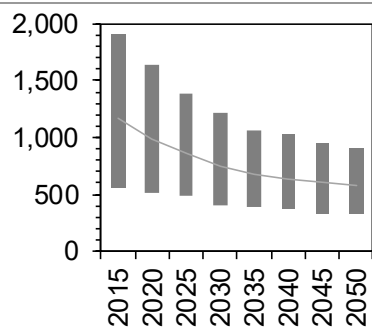




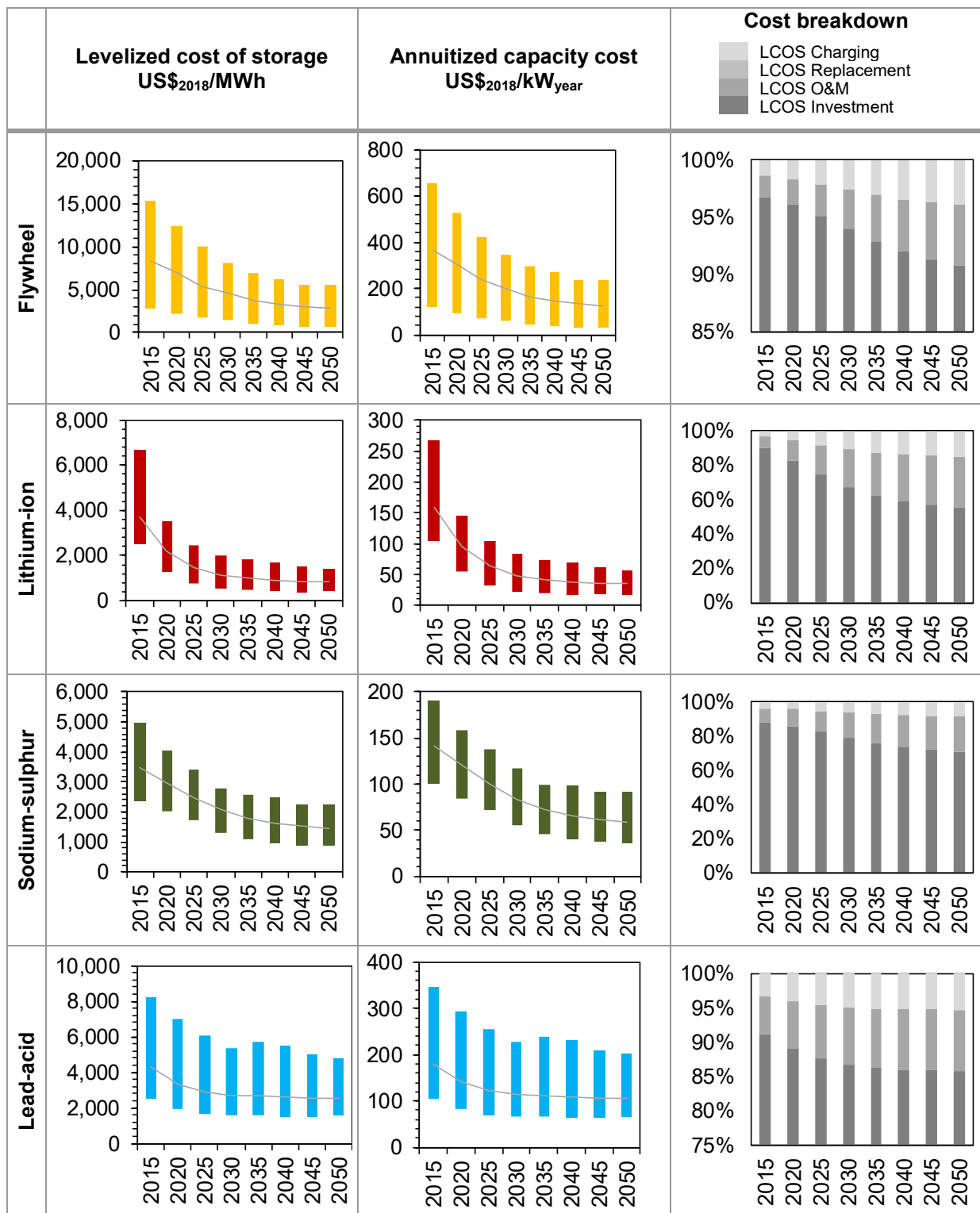
Application: Bill Management

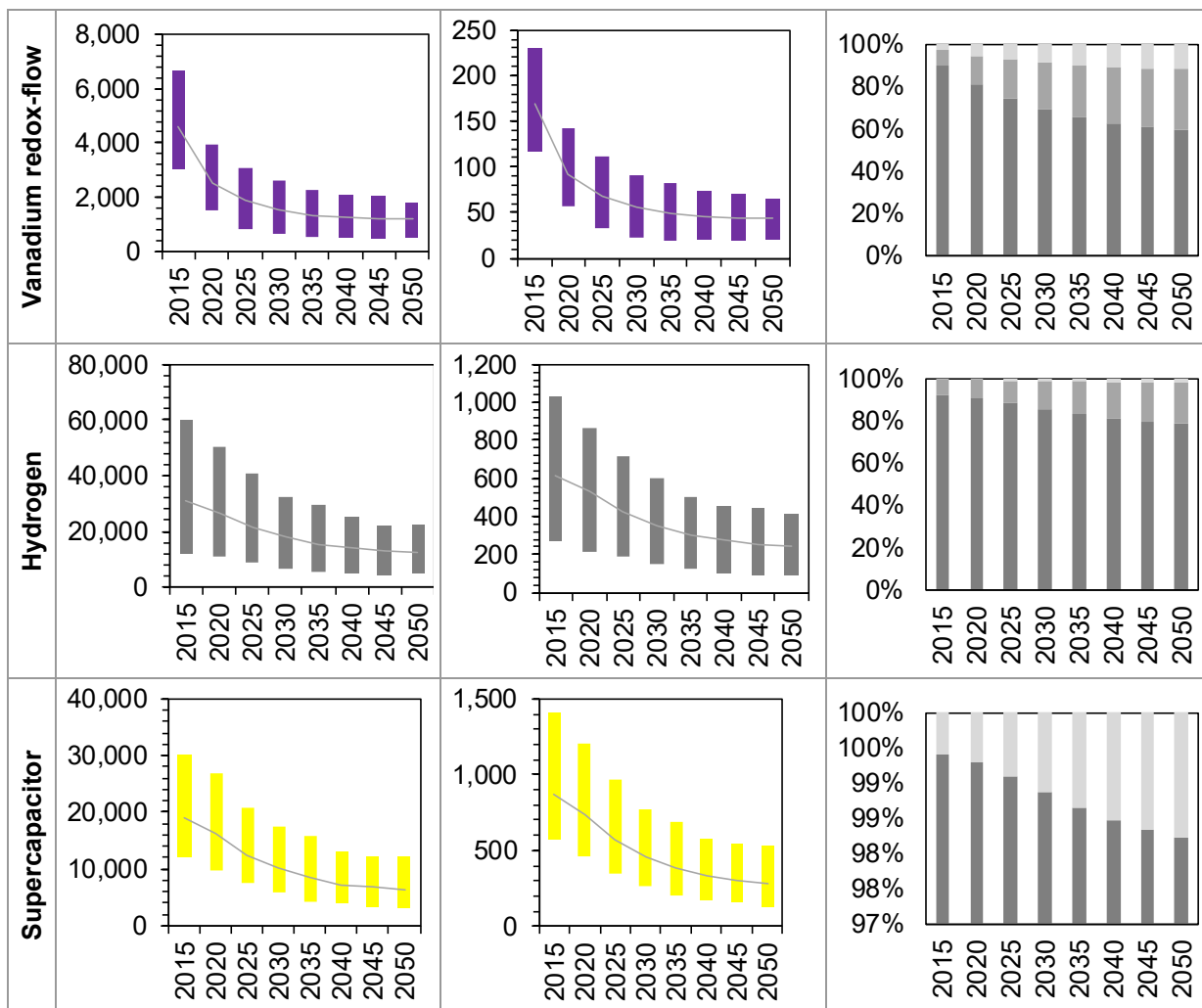


Hydrogen

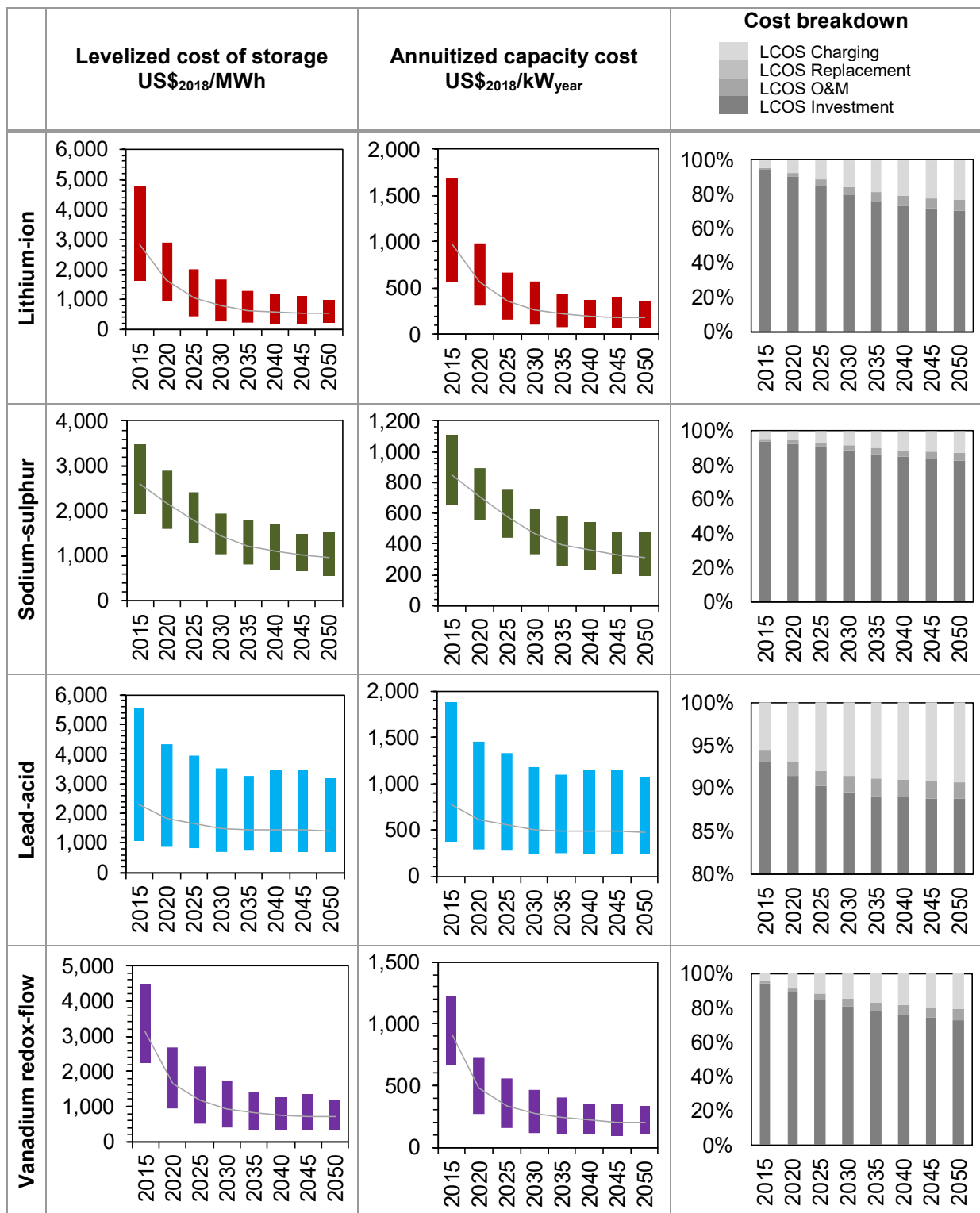


Application: Power Quality





Application: Power Reliability



Hydrogen

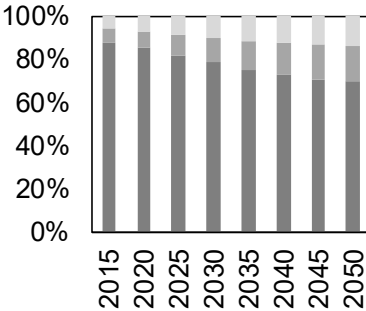
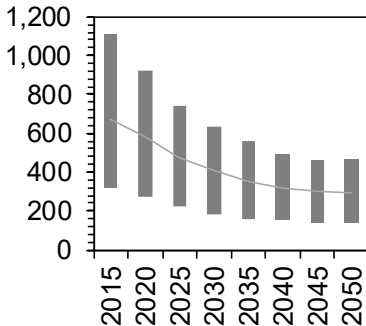
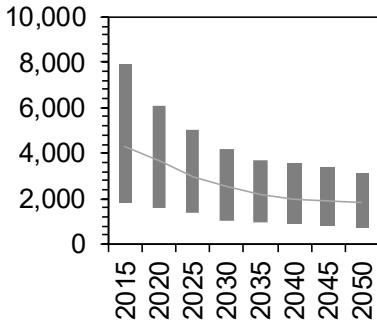


Figure S4. Overview of technology probabilities to exhibit lowest LCOS and their mean LCOS in power terms or annuitized capacity cost

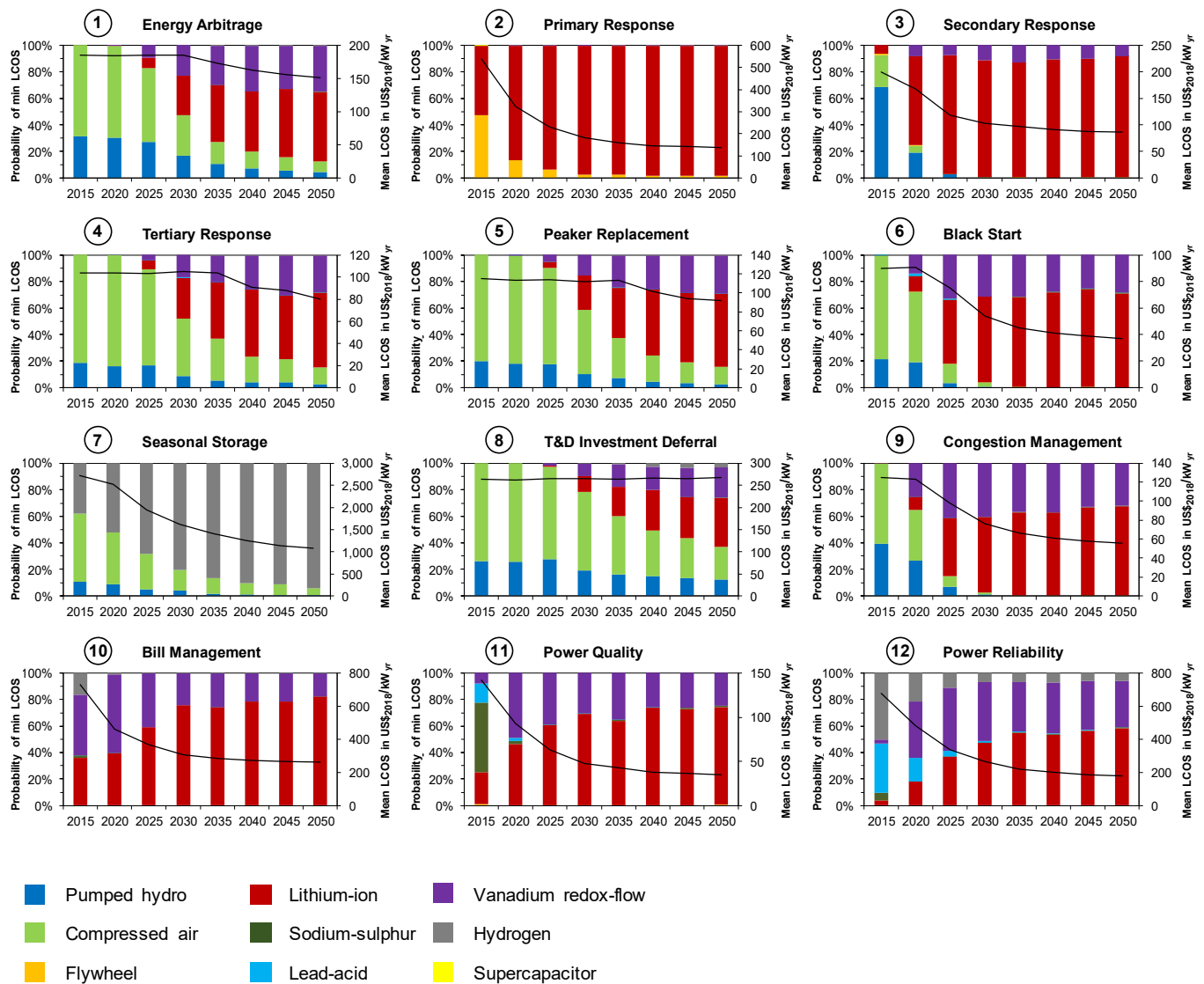
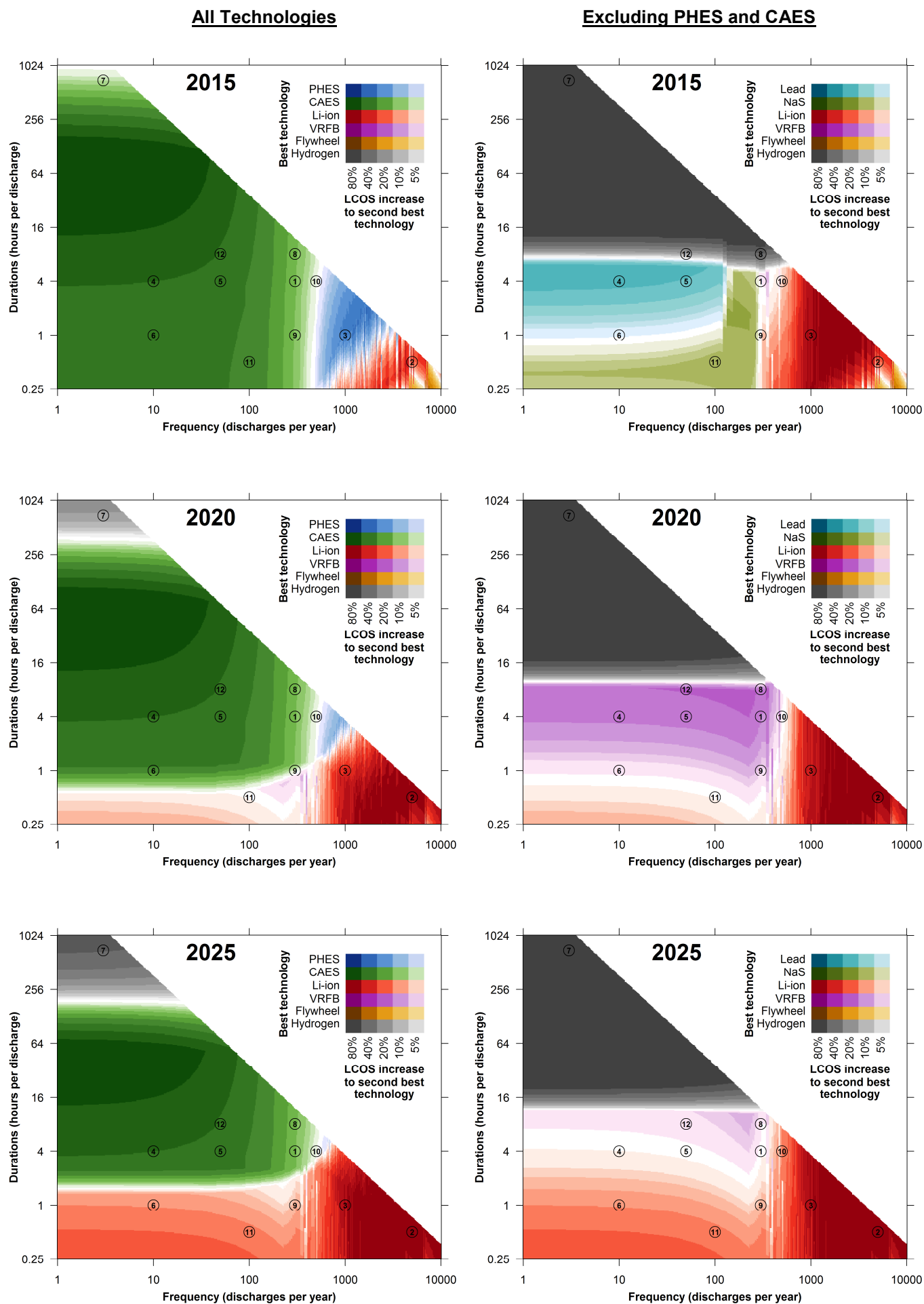


Figure S5. Lowest annuitized capacity cost (US\$/kW_{year})



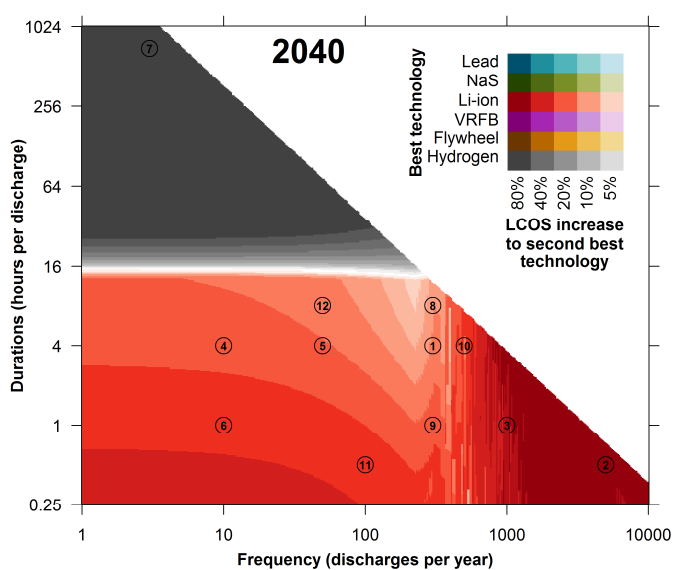
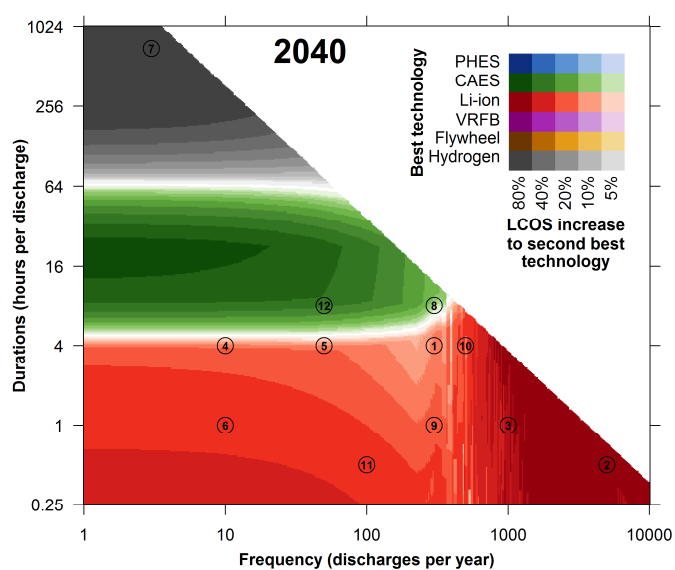
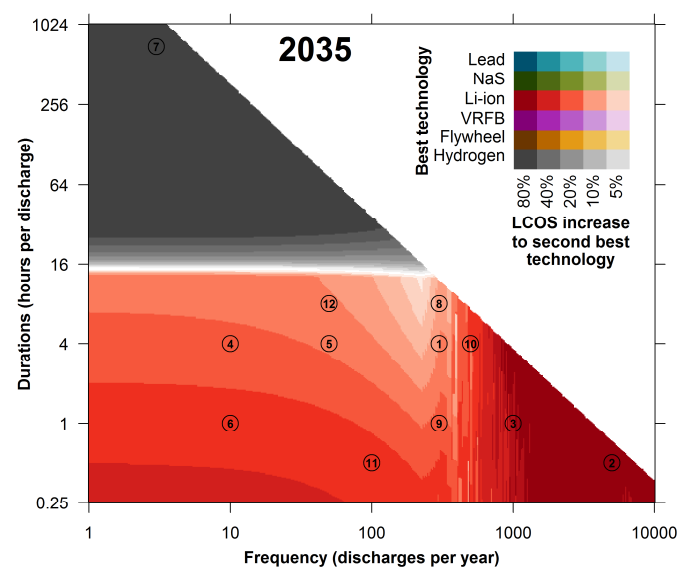
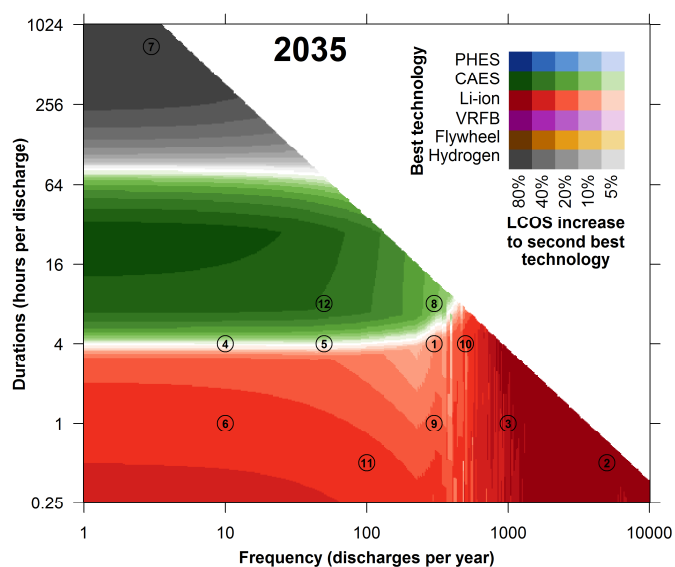
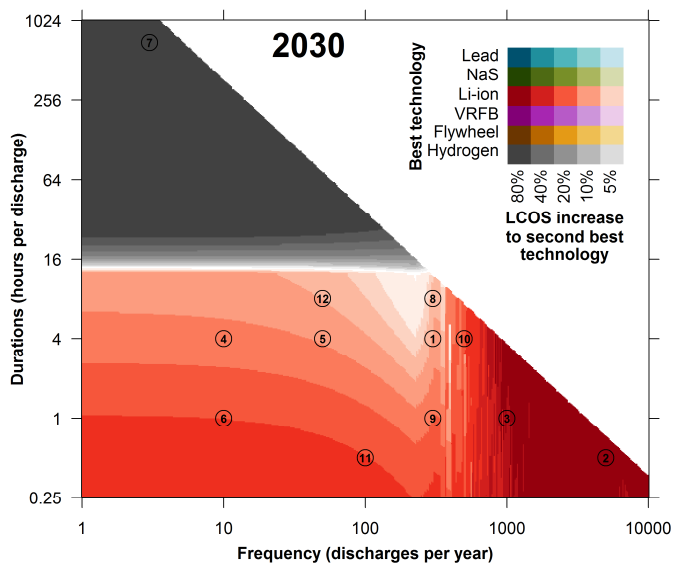
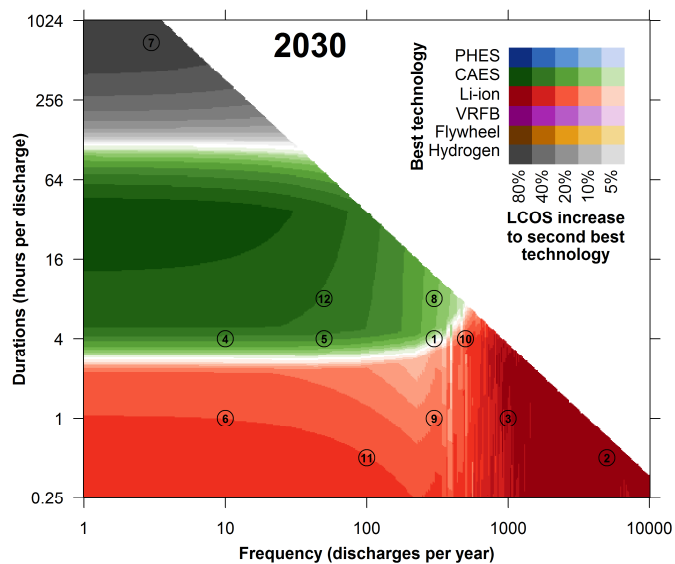


Figure S6. Annuitized capacity cost (US\$/kW_{year}) – Most cost-efficient technology

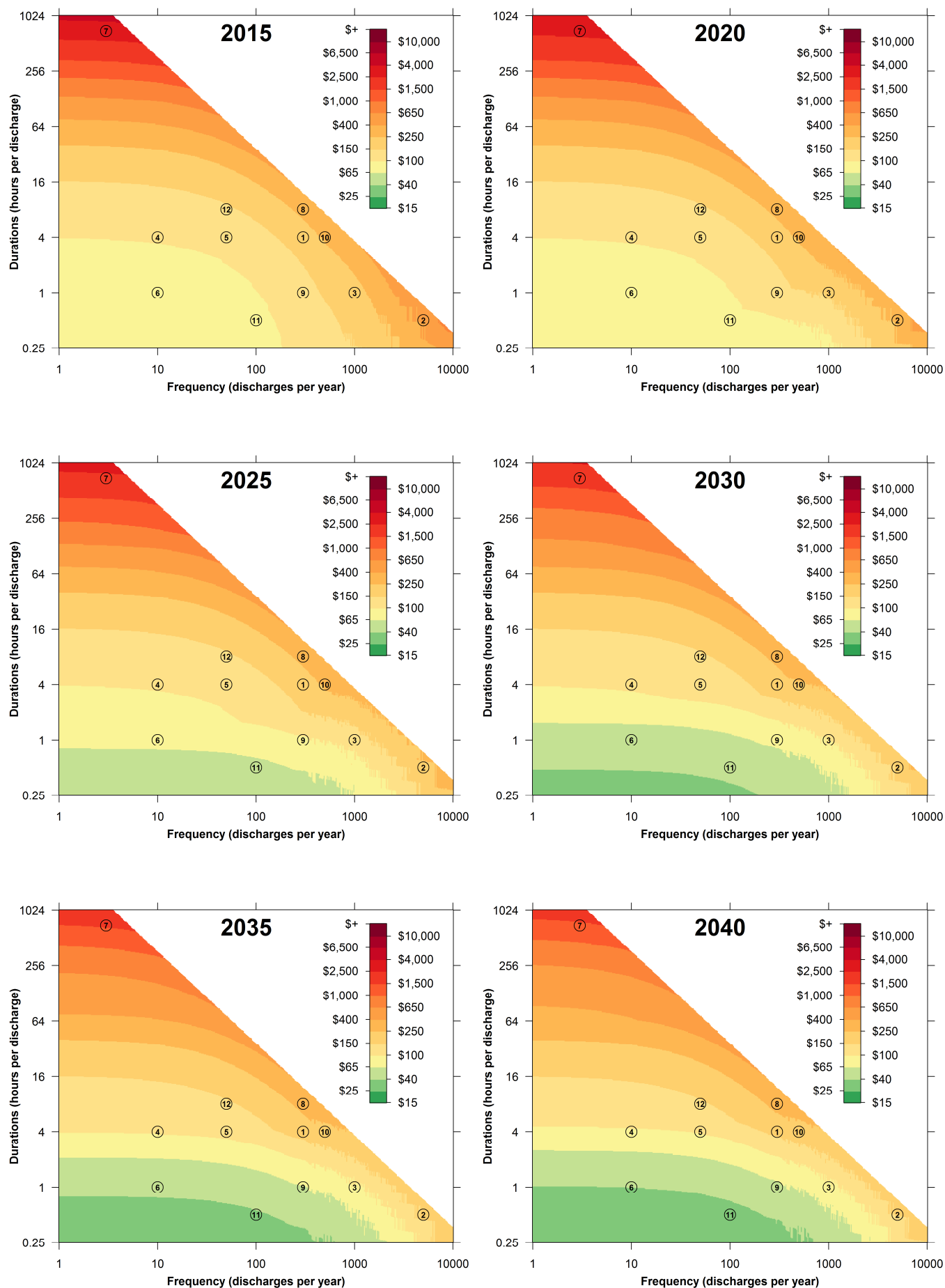


Figure S7. Sensitivity to power price

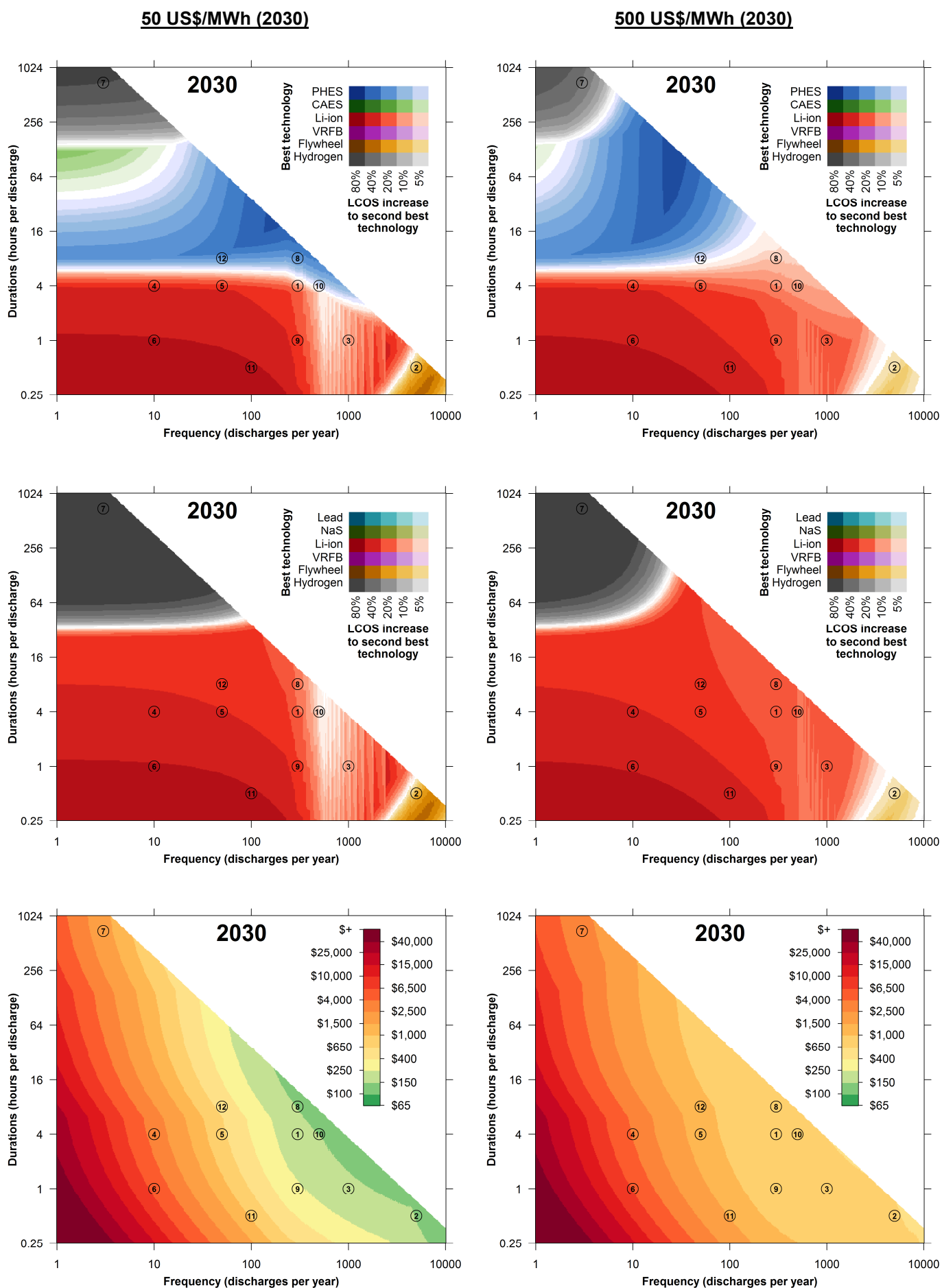


Figure S8. Sensitivity to discount rate

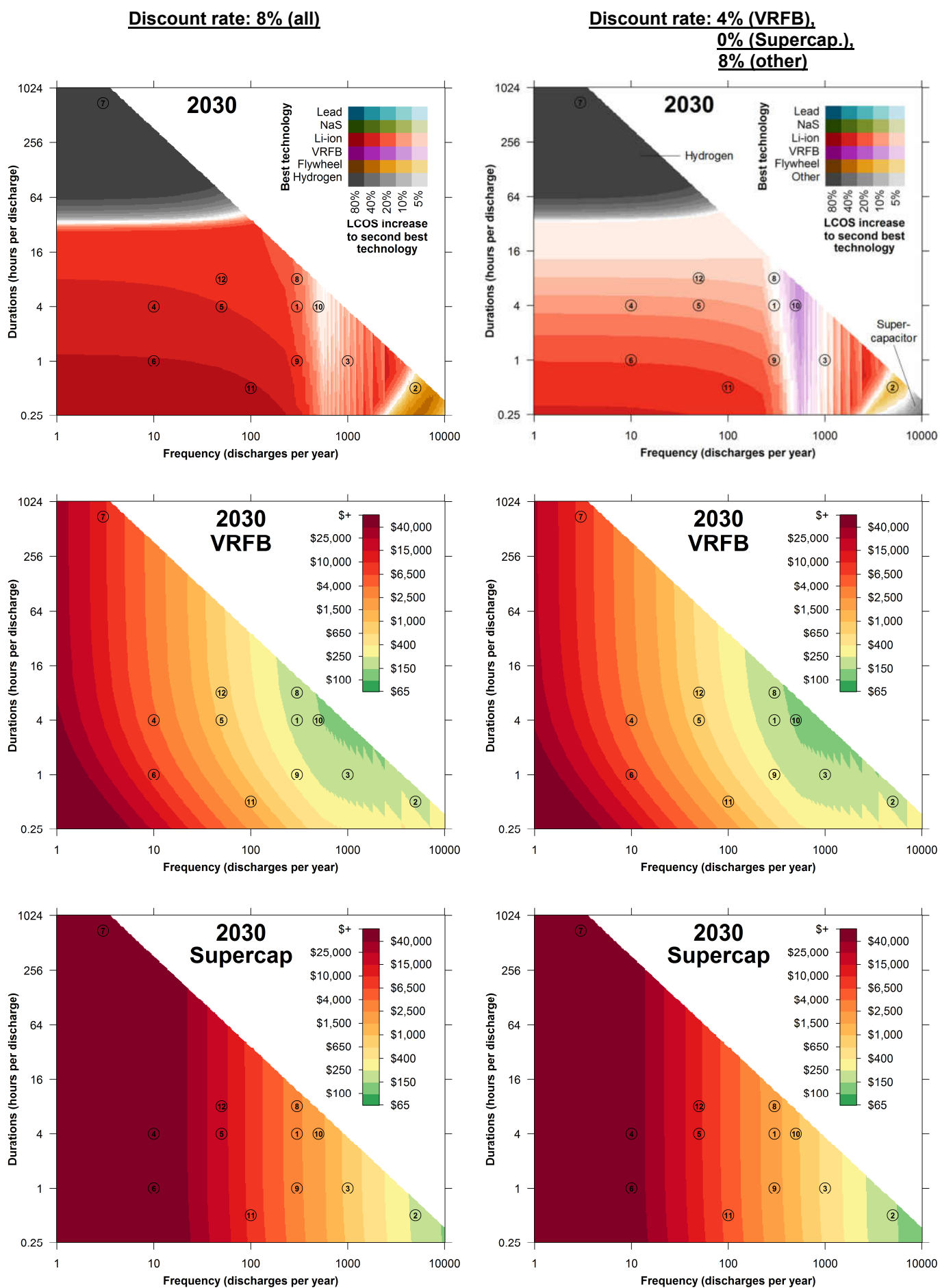
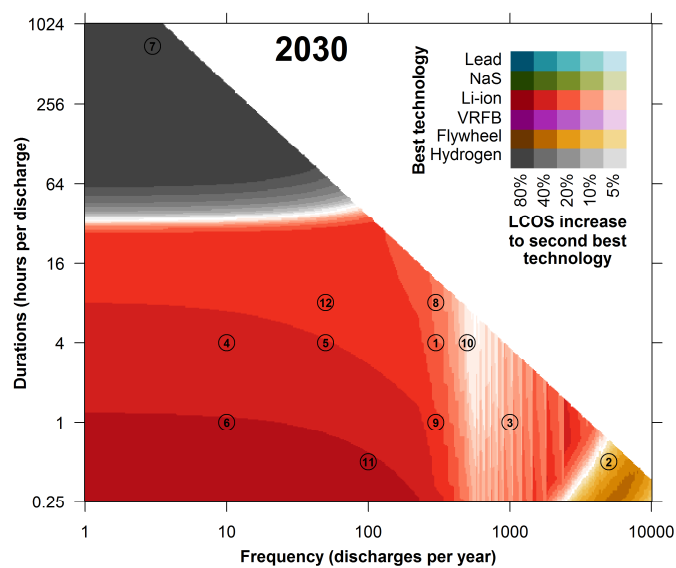


Figure S9. Sensitivity of vanadium redox-flow LCOS to efficiency improvement

Vanadium redox-flow – Round-trip efficiency: 73%



Vanadium redox-flow – Round-trip efficiency: 85%

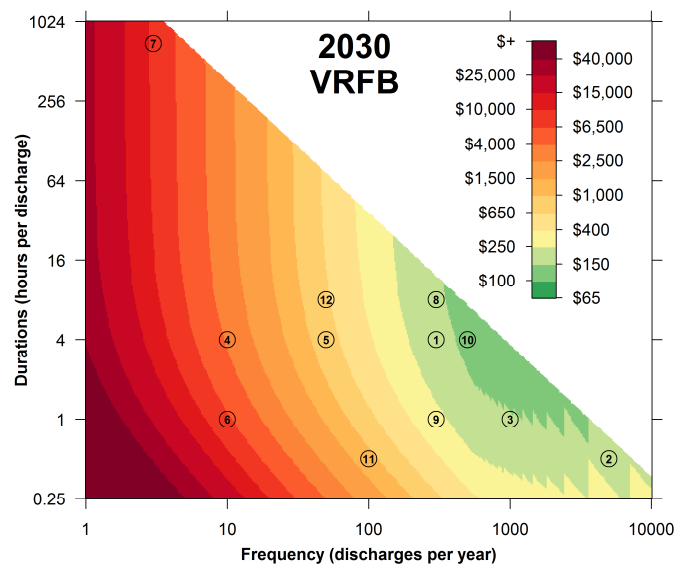
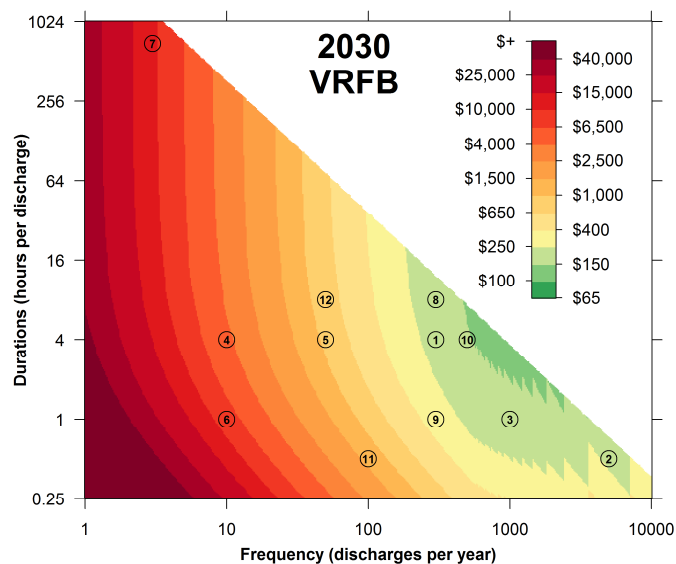
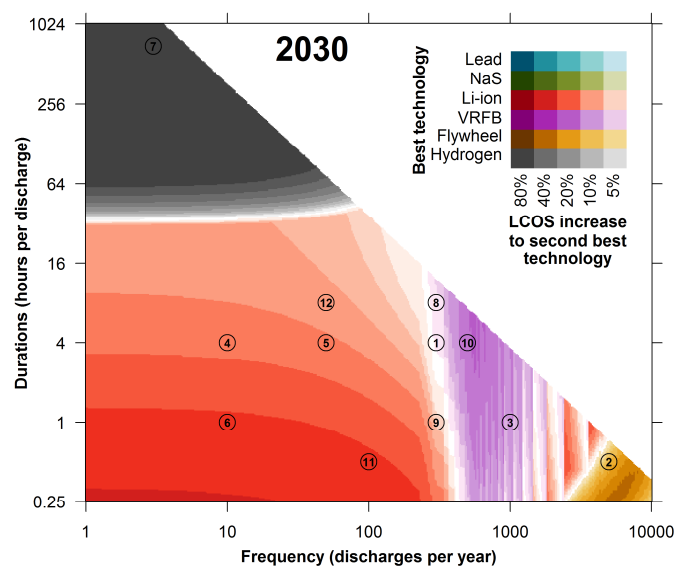


Figure S10. Sensitivity of vanadium redox-flow LCOS to lifetime improvement

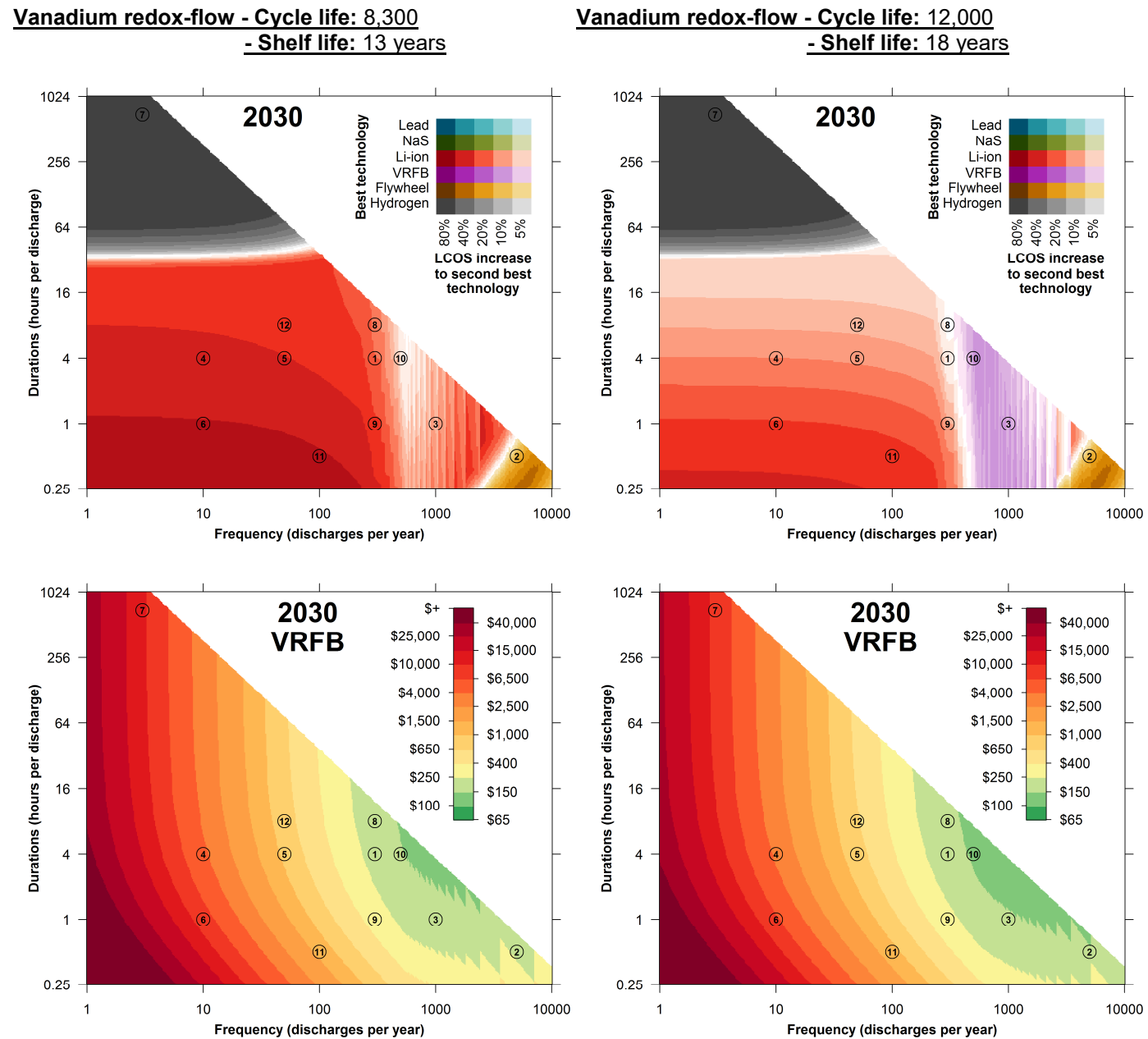
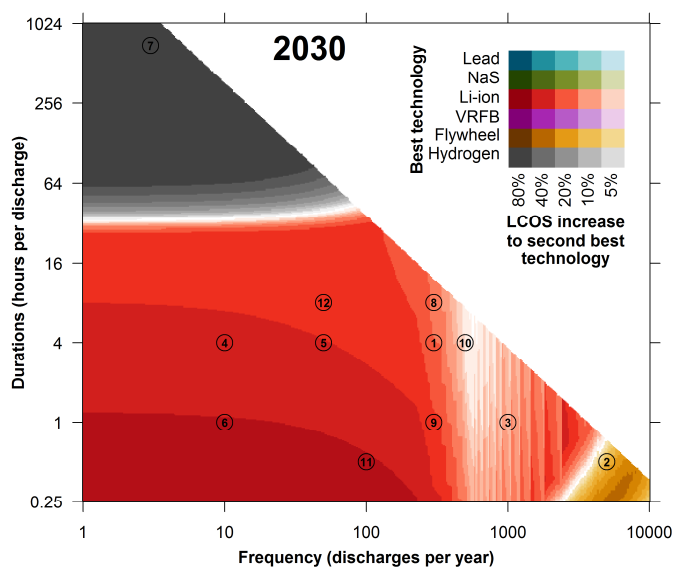


Figure S11. Sensitivity of lead-acid LCOS to efficiency and lifetime improvement

Lead-acid - Round-trip efficiency: 84%

- Cycle life: 1,200

- Shelf life: 10 years



Lead-acid - Round-trip efficiency: 95%

- Cycle life: 5,000

- Shelf life: 40 years

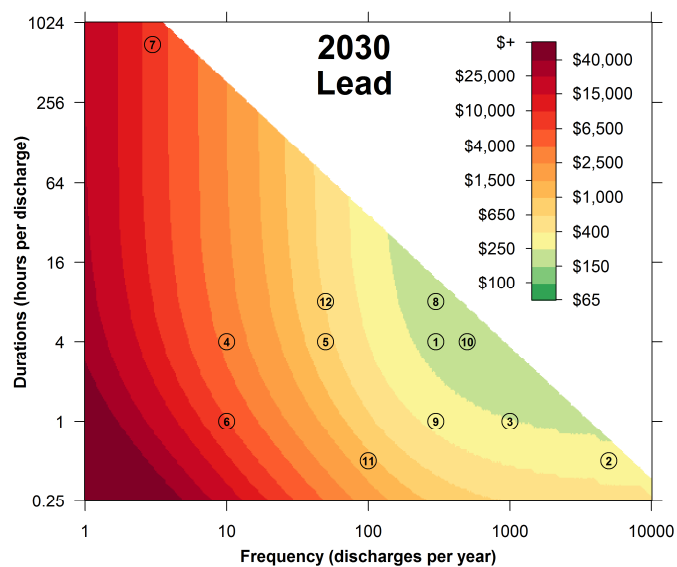
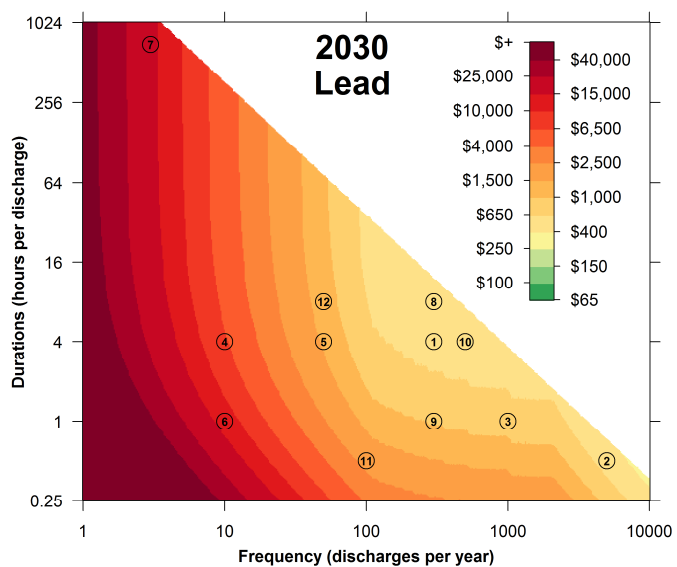
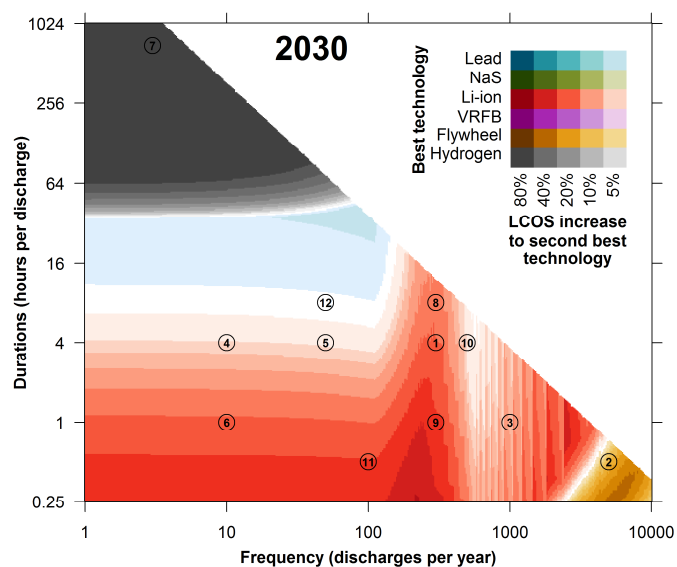
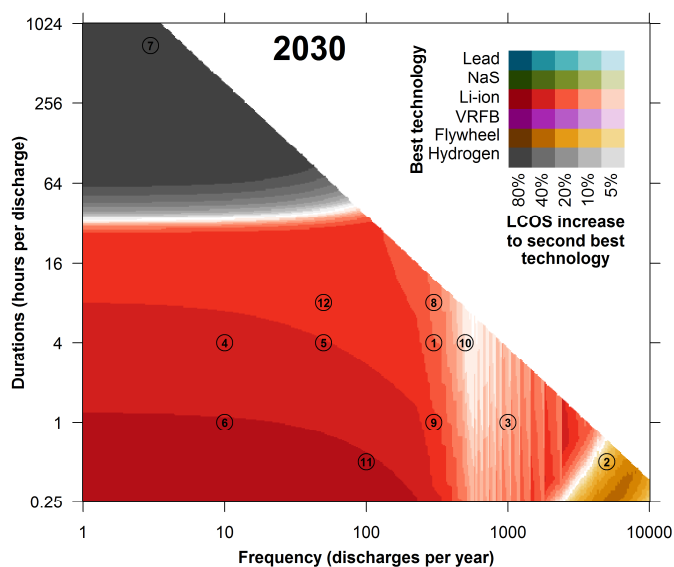


Figure S12. Sensitivity of sodium-sulphur LCOS to efficiency and lifetime improvement

Sodium-sulphur - Round-trip efficiency: 81%

- Cycle life: 4,000

- Shelf life: 14 years



Sodium-sulphur - Round-trip efficiency: 94%

- Cycle life: 17,000

- Shelf life: 20 years

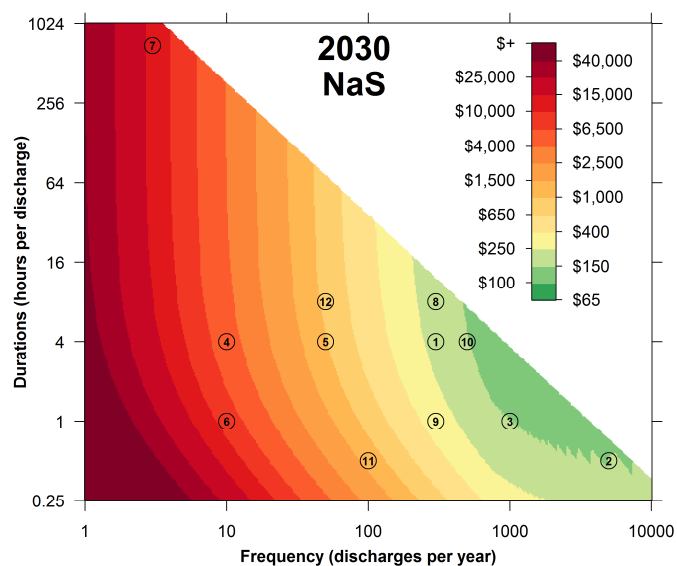
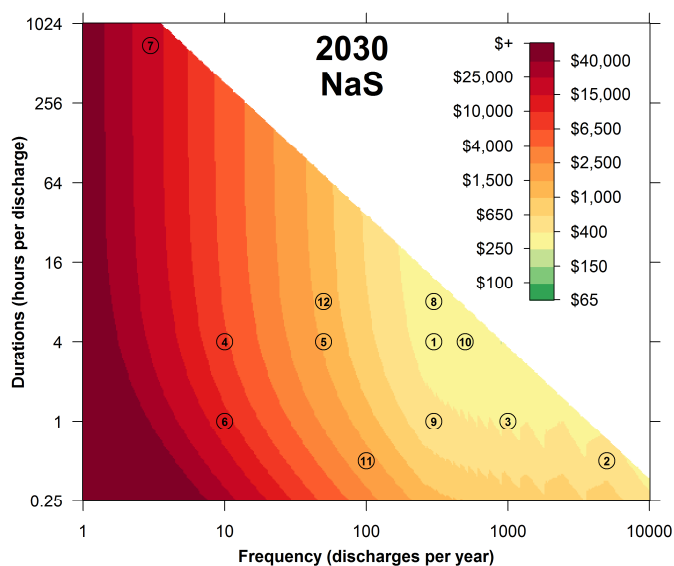
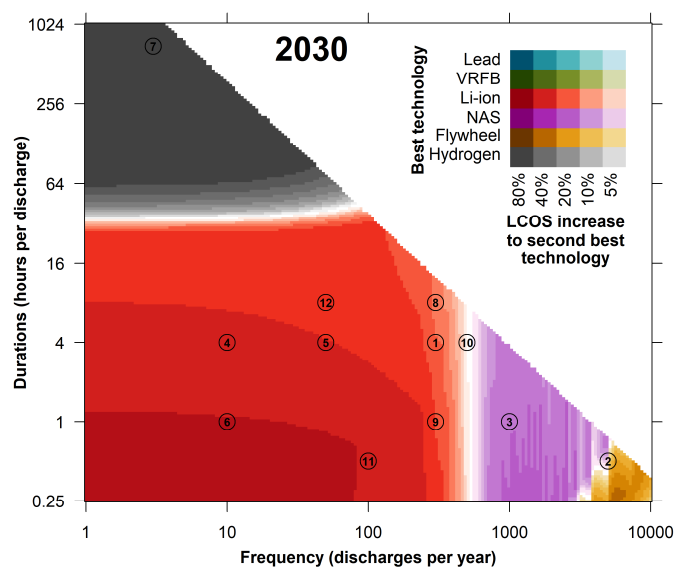


Figure S13. Sensitivity of supercapacitor LCOS to efficiency and lifetime improvement

Supercapacitor - Round-trip efficiency: 91%

- Cycle life: 300,000

- Shelf life: 14 years

Supercapacitor - Round-trip efficiency: 96%

- Cycle life: 600,000

- Shelf life: 58 years

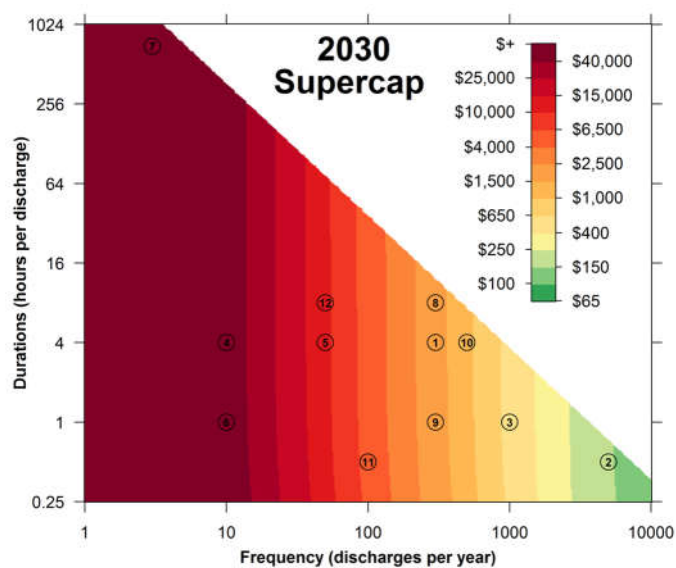
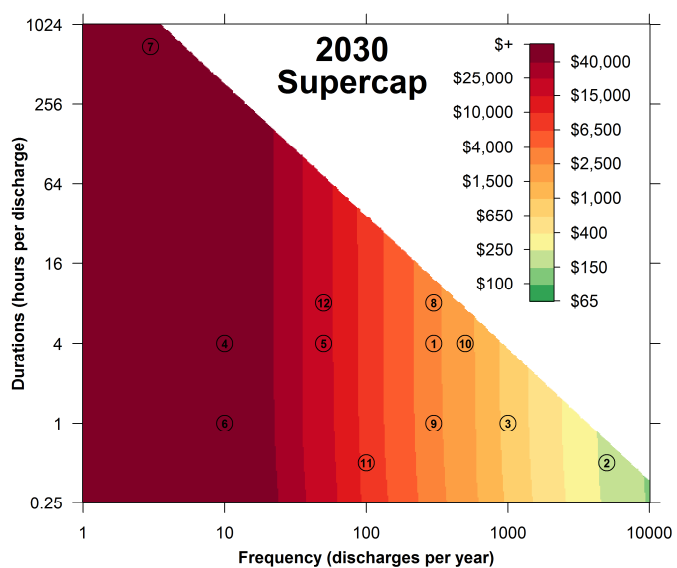
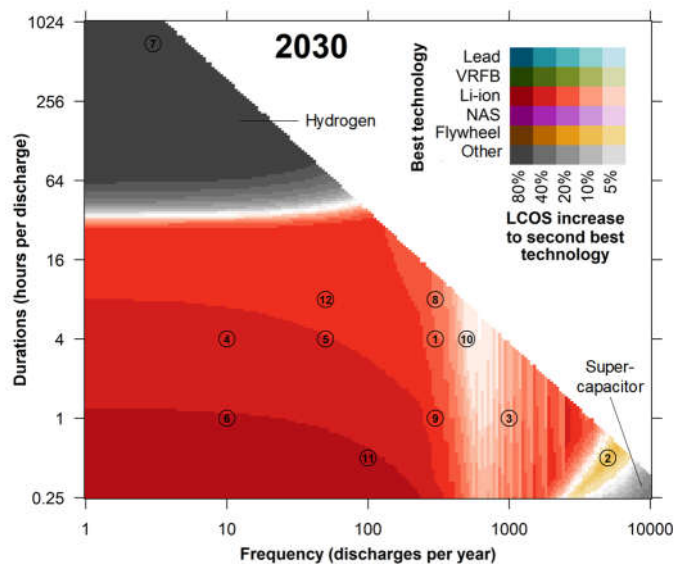
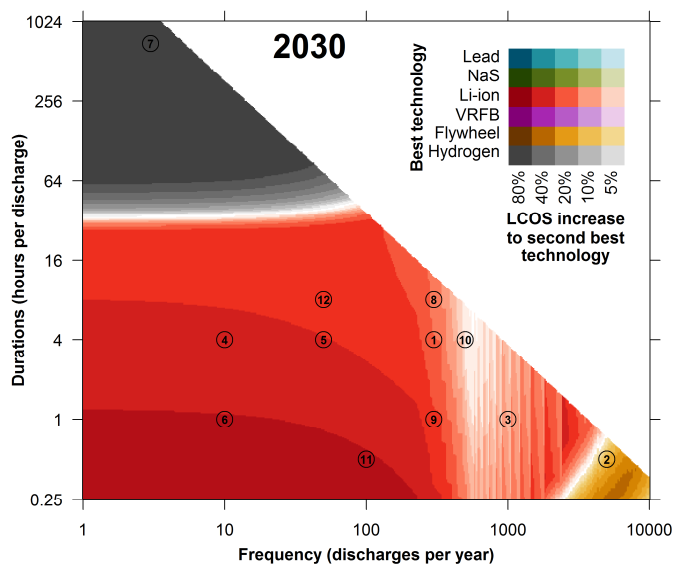


Figure S14. Result comparison to other LCOS studies for Energy Arbitrage

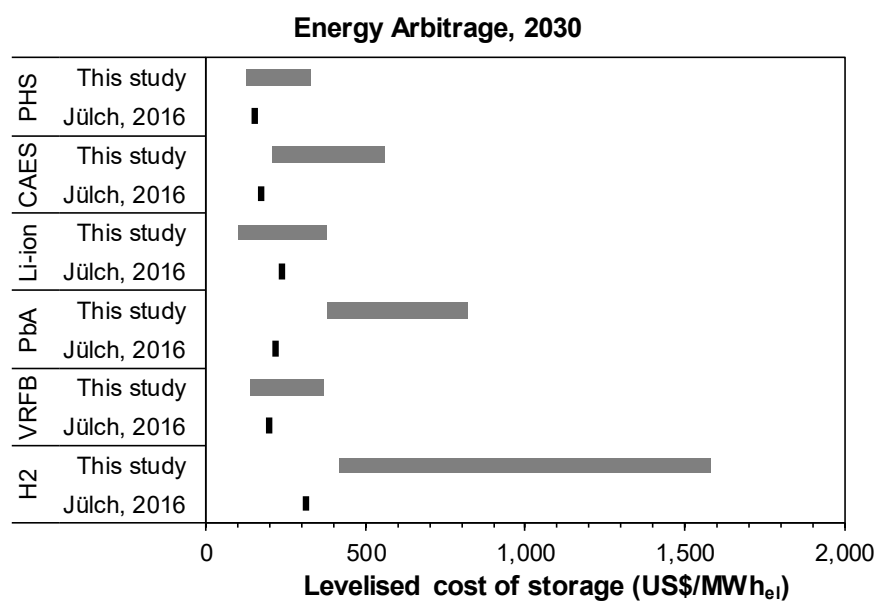
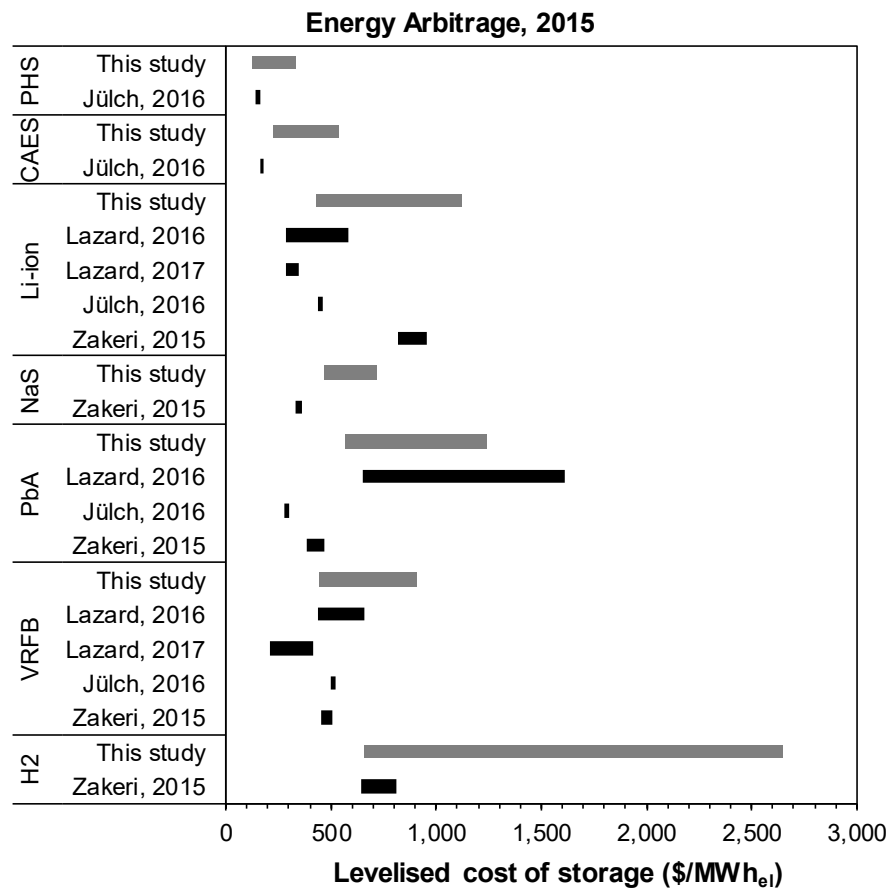


Figure S15. Result comparison to other LCOS studies for Primary Response

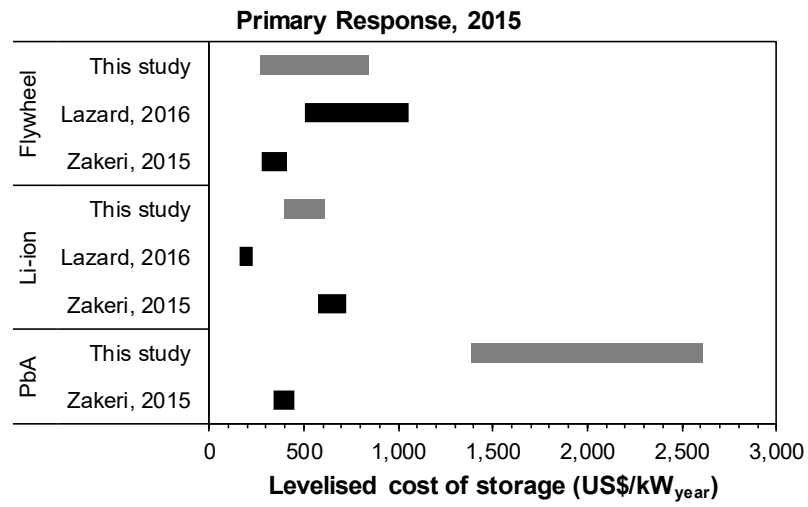


Figure S16. Result comparison to other LCOS studies for Seasonal Storage

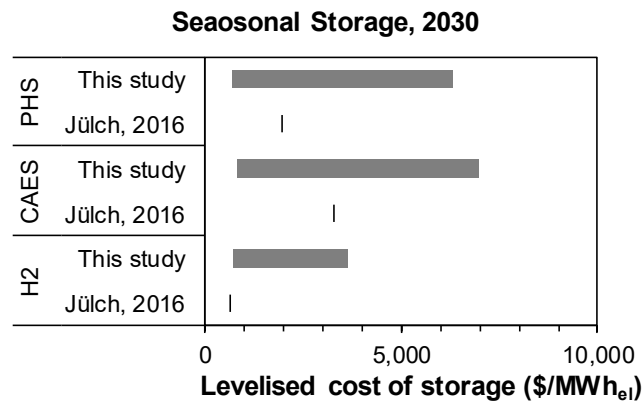
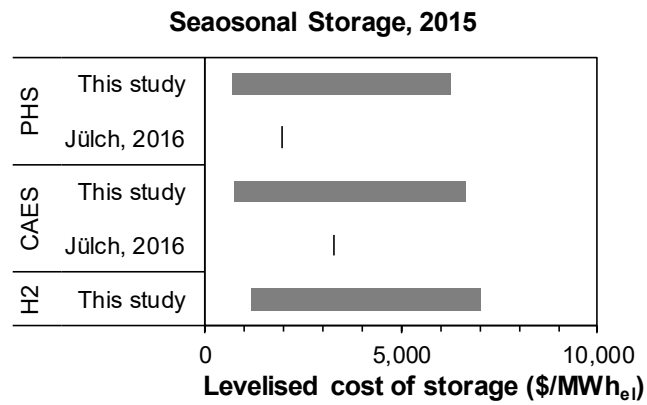
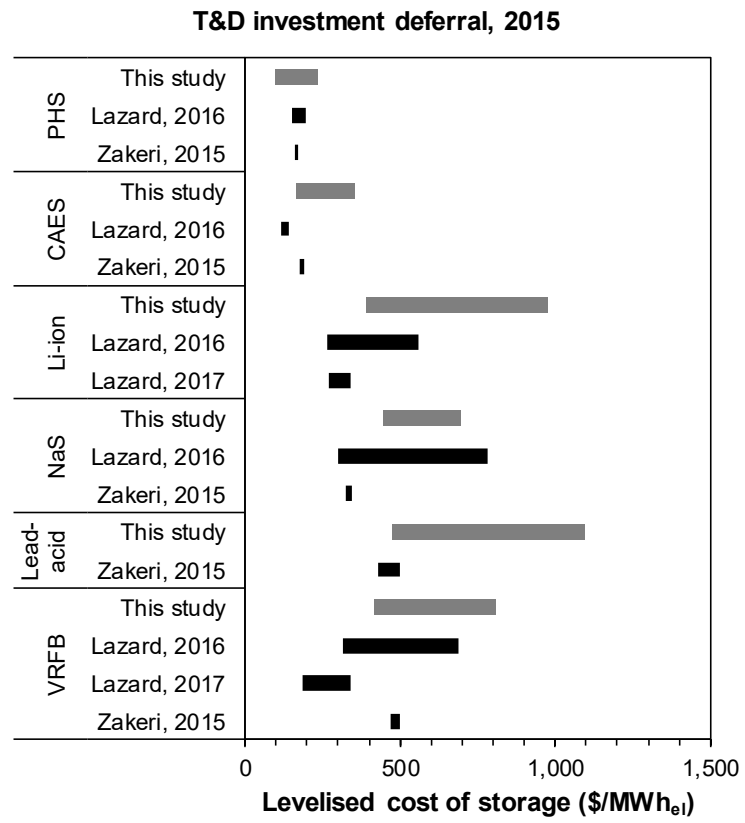


Figure S17. Result comparison to other LCOS studies for T&D investment deferral



Supplemental Experimental Procedures

Procedure S1. Methodological LCOS comparison to recent studies

Recent LCOS studies differ with respect to parameters included in the calculation and their methodological approach.

Investment costs should include overnight investment cost, construction time and replacement cost. While most studies consider overnight investment and replacement cost, they neglect the impact of construction time. In this study, the discounting impact of construction time is accounted for in investment cost, as well as operating and charging costs and electricity discharged. Also, most studies combine operating and replacement cost and thereby annualise replacement cost, which distorts the impact of discounting on this cost element. In this study, replacement costs are fully discounted to the end of each replacement period.

The end-of life cost or potential value of a storage technology at its end of life or end of investment period should also be accounted for, which is not the case for some studies.

Electricity discharged incorporates the technologies nominal charge capacity, depth-of-discharge, efficiency, annual cycles, self-discharge and time and cycle degradation. Some studies neglect the impact of varying depth-of-discharge and most studies neglect the impact of self-discharge and degradation.

Some studies do not explicitly consider the lifetime of the technology, but rather assume a fixed investment period. This approach is suitable for specific project proposals, but ill-suited to compare technologies based on their inherent cost and performance characteristics.

Considering the impact of taxes is important in a country-specific context, but not relevant for location-independent character of this study.

	LCOS components	Zakeri et al. ⁷	Jülch et al. ³⁰	Lazard ^{31,32}	Lai et al. ³³	Pawel ³⁴	Battk et al. ³⁵	This study
Economic	Investment cost	x	x	x	x	x	x	x
	Replacement cost	x	x	x				x
	Operating cost	x	x	x	x	x	x	x
	Power cost	x	x	x	x	x	x	x
	End-of-life cost	x	x			x		x
	Discount rate	x	x	x	x	x	x	x
	Taxes			x				
Technical								
	Nominal capacity	x	x	x	x	x	x	x
	Depth of discharge	x	x	x		x	x	x
	Round-trip efficiency	x	x	x	x	x	x	x
	Cycle life	x	x				x	x
	Shelf life	x	x		x	x	x	x
	Construction time							x
	Degradation rate				x	x		x
	Self-discharge		x					x

In terms of methodological approach, Lai et al.³³ differentiate between the levelized cost of storage, excluding power price and round-trip efficiency, and levelized cost of delivery, including both. All other studies include power price and round-trip efficiency in their lifetime cost assessment of storage technologies, and three of them call this metric levelized cost of storage^{30–32}. In this study *levelized cost of storage* is chosen as metric including power price and round-trip efficiency, because the nomenclature is meaningful and different to *levelized cost of electricity*, which is used for generation technologies, and a metric excluding technology round-trip efficiency is considered incomplete.

Procedure S2. Comparison of LCOS studies

Multiple recent academic and industry studies perform LCOS analyses for different electricity storage technologies and applications^{7,30–32}. Figures S8 to S11 compare the findings of these studies to the present analysis.

Energy Arbitrage (Discharge duration: 4 hours, Annual cycles: 300)

For energy arbitrage, the LCOS values identified by Jülch et al.³⁰ for “short-term storage” (4h discharge, 365 cycles) are within the ranges identified in this study, although often at the lower end, and for lead-acid batteries significantly below. These deviations have two main reasons. First, annual cycle assumptions differ (365 vs 300). This effect reduces LCOS by 8-15%. More significantly, distinct cycle life assumptions are higher and investment cost assumptions are lower than in the present study (lithium-ion: 7,000 vs 3,250 cycles, lead-acid: 2,700 vs 1,250 cycles; pumped hydro and compressed air: 50% lower investment cost). This also explains why Jülch et al.’s results for 2030 are at the lower end of the ranges identified in this study. Exceptions are lithium-ion and hydrogen storage. Despite the higher cycle life assumption (10,000 vs 3,250), respective LCOS are at the upper end of our range, because of Jülch et al.’s higher investment cost assumptions for 2030. For hydrogen, respective LCOS are below the range identified in this study due to lower investment cost and longer shelf life assumptions. The deviations in investment cost and lifetime assumptions used by Jülch et al. could not be verified in the academic and industry literature used for this study.

In the two LCOS studies produced by Lazard^{31,32}, the peaker replacement application features similar requirements that we model for energy arbitrage (4h, 365 cycles). The LCOS values are also at the lower end of the results in the present study (Lazard, 2016) or even lower (Lazard, 2017). Again, the slight difference in annual cycle requirement explains part of the deviation. Lower charging cost (30 US\$/MWh vs 50 US\$/MWh) also has an impact. However, the most significant deviations are found for lithium-ion and vanadium redox-flow due to the significantly lower investment cost assumptions. This can partly be explained with the temporal difference of the studies. Lazard’s investment cost values are for 2016 or 2017 and assessed against the 2015 values of this study. Especially these two technologies have experienced significant cost reductions in recent years (Figure S2).

The LCOS values identified by Zakeri et al.⁷, are well within the ranges of the present study, apart from sodium-sulphur and lead-acid batteries. For sodium-sulphur this is due to lower investment cost, and for lead-acid due to higher cycle life assumptions. Both could not be verified with recent industry reports. Zakeri et al.’s range identified for hydrogen storage is at the lower end of our range due to investment cost assumptions at the lower end compared to this study.

Primary Response (Discharge duration: 0.5 hours, Annual cycles: 5,000)

Lazard³¹ and Zakeri et al.⁷ also model LCOS for flywheels, lithium-ion and lead-acid batteries for a primary response application, albeit significantly different annual cycle requirements (500 vs 1,000 vs 5,000 cycles). LCOS in power terms or annuitized capacity costs value the provision of capacity instead of discharged electricity. As such, an increase in discharged electricity is not valued and many annual cycles only limit the technology’s lifetime. As a result, LCOS are lowest for the study assuming fewest annual cycles (Lazard and Zakeri et al.). Deviations beyond this effect are the result of differing lifetime assumptions (lead-acid, Zakeri et al.; flywheels, Lazard).

Seasonal storage (Discharge duration: 700 hours, Annual cycles: 3)

LCOS for seasonal storage were also modelled by Jülch et al.³⁰. For pumped hydro and compressed air, the results match with the ranges identified in this study. This is the result of two contrasting effects. Jülch et al. only assume 1 annual cycle, which would mean LCOS should be around 3 times higher than in this study. However, the 50% lower investment cost assumptions for pumped hydro and compressed air and longer lifetime assumptions mean that LCOS fall back into the range identified in this study. For hydrogen storage, the relevant energy-specific investment cost in Jülch et al. are 50 times lower, which explains their LCOS result below the range in this study.

T&D investment deferral (Discharge duration: 8 hours, Annual cycles: 300)

Despite slightly differing discharge duration and annual cycle requirements, the LCOS results of T&D investment deferral are compared to the LCOS results of Lazard 2016³¹ (application: Transmission), Lazard 2017³² (application: Distribution substation) and Zakeri et al.⁷ (application: Bulk storage). For most technologies there is broad agreement in the LCOS identified in all studies.

An exception is compressed air due to the high efficiency assumption made by Lazard 2016. The low LCOS for lithium-ion and vanadium redox-flow by Lazard are again the result of significantly lower investment cost assumptions, which can be the result of the temporal difference of the studies, given recent cost reductions for both technologies (see above).

Zakeri et al.'s results are lower for lead-acid and sodium-sulphur, which is the result of lower investment cost assumptions. Again, these investment costs could not be verified in the academic and industry literature used for the present study.

In addition, the LCOS ranges identified in this study tend to be slightly higher than the results in the discussed studies due to the differences in LCOS modelling approach (Note S1). The neglect of construction time, capacity degradation and self-discharge artificially reduces LCOS in these studies. However, the effect is marginal compared to the impact of the described differences in investment cost and lifetime assumptions.

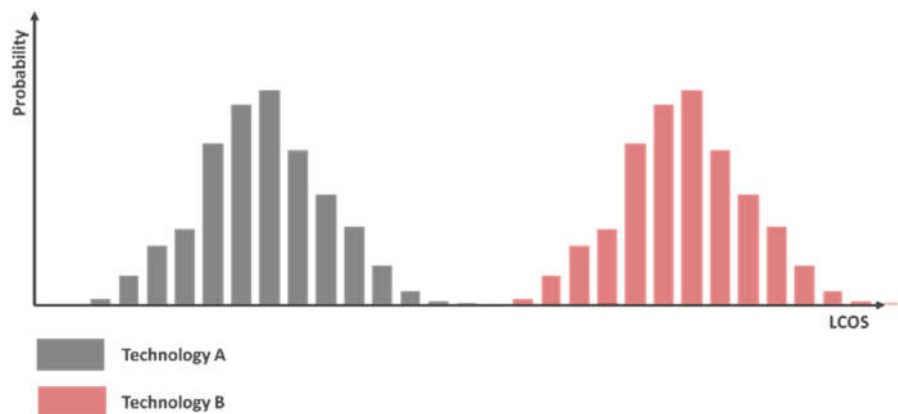
Procedure S3. Probability assessment

The Monte-Carlo simulation of the LCOS calculation gives 500 LCOS results for each modelled technology in a specific application. The LCOS results are normally distributed due to the normal distribution associated with the input parameters.

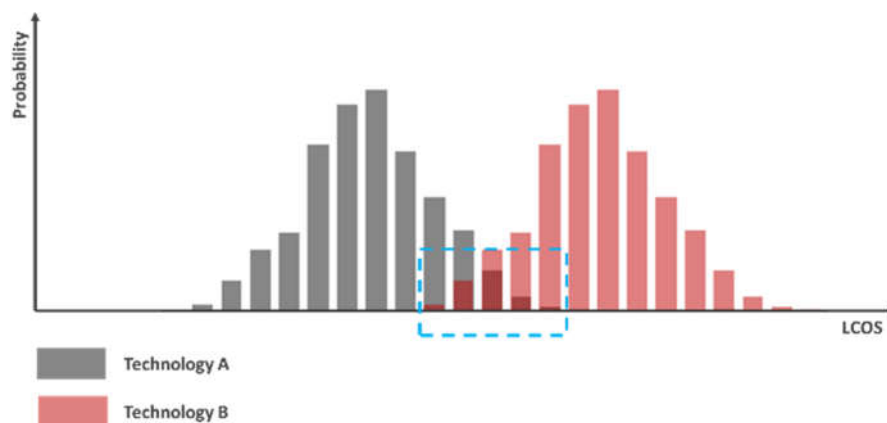
If the maximum of the LCOS distribution for a technology A is below the minimum of all the other technologies, the simulation sets the technology A as the cheapest option with 100% probability (Option 1).

However, if the intersection between the LCOS distributions is not an empty set, the probability is lower (Option 2). The approach then counts the occurrences when technology A exhibits lower LCOS than all other technologies and divides by all occurrences ($500^{\text{\#_technologies}}$) to arrive at the probability for a technology to exhibit lowest LCOS.

Option 1: 100% probability



Option 2: <100% probability



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